

AForce — Complete Specifications

Generated 2026-06-23

- [replit.md](#)
- [Overview](#)
- [FINAL BUILD LOCK](#) (locked; do not redesign)
 - [AUTHORIZED OVERRIDE — Home + Navigation redesign](#) (owner-approved)
 - [Cold-Launch Opening Sequence](#) (owner-approved)
 - [Water-First Command System](#)
 - [Score Protection Rule](#)
 - [Language / Localization Lock](#)
 - [Engine / UI Governance](#)
 - [Product Positioning Rule](#)
 - [MVP Surfaces](#) (do not remove)
- [User Preferences](#)
- [System Architecture](#)
 - [Core Technologies](#)
 - [UI/UX Decisions](#)
 - [Authentication & Identity](#)
 - [Subscription & Entitlement](#)
 - [Publishing & Distribution](#)
 - [Release-Readiness Status](#) (verified)
 - [Server Hardening](#)
 - [Technical Implementations & Feature Specifications](#)
 - [Architecture Diagram](#)
- [External Dependencies](#)
- [artifacts/aforce-os/docs/AForce-OS-Specification.md](#)
- [AForce OS — Product & Technical Specification](#)
 - [1. Product Overview](#)
 - [2. Product Locks](#) (non-negotiable design rules)
 - [3. Technical Architecture](#)
 - [4. App Surfaces — Screens & Navigation](#)
 - [5. Feature Flags & Phased Exposure](#)
 - [6. Hydration Scoring Engine](#)
 - [7. Performance-State Color System](#)
 - [8. Subscription Tiers](#)
 - [9. Product Store](#) (Physical SKUs)
 - [10. Data Model](#) (PostgreSQL / Drizzle)
 - [11. API Surface](#) (`api-server`)
 - [12. Integrations](#)
 - [13. Localization](#)
 - [14. Publishing & Distribution](#)
 - [15. Design System](#) (Brand v2.1.0)
- [artifacts/aforce-os/SPEC-SHEET.md](#)
- [AForce OS — Product & Technical Specification](#)
 - [At a Glance](#)
 - [1. Platform & Technology Stack](#)
 - [2. Application Architecture](#)
 - [3. Navigation & Screen Inventory](#)
 - [4. Onboarding & Cold-Launch Opening Sequence](#)
 - [5. Hydration Scoring Engine](#)
 - [6. Recovery & Health Signals](#)
 - [7. AI Coach / Voice Engine](#)

- 8. Social Layer
- 9. Store & Subscription System
- 10. Authentication & Identity
- 11. Design System
- 12. Localization
- 13. Feature Flags
- 14. Backend & API
- 15. Build, Distribution & Permissions
- Appendix — Key Dependency Versions
- [\[\] AFORCE_FINAL_SPEC.md](#)
- AForce OS — Final Spec (Core Product)
 - Implementation Order
 - Hard Rules
 - Final Refinement Locks
 - Phase 10 — Investor Demo Overlay
 - Product Surface (high-level)
 - Design Language
 - Out of Scope (this document)
- [\[\] AFORCE_PHASE_STATUS.md](#)
- AForce OS — Phase Status
 - Core Phases (AFORCE_FINAL_SPEC.md)
 - Addon Phases (AFORCE_SOCIAL_CRUISE_ADDON.md)
- [\[\] AFORCE_SOCIAL_CRUISE_ADDON.md](#)
- AForce OS — Social + Cruise Enhancement Layer (Addon)
 - Activation Order
 - Social Additions
 - Cruise Additions
 - Explicitly Out of Scope (never build under this addon)
 - Hard Rules
- [\[\] design/aforce-design-tokens.md](#)
- AForce OS Design Tokens — AForce Brand System
 - Design Philosophy
 - 1. Colors
 - 2. Typography (Archivo Black · IBM Plex Mono · Inter)
 - 3. Spacing
 - 4. Radii
 - 5. Component Dimensions
 - 6. iPhone 14 Pro Layout
 - 7. Shadows / Glows
 - How to Use in Figma
- [\[\] artifacts/aforce-os/docs/social-mode-safety-spec.md](#)
- Social Mode — Safety & Estimation Spec
 - 1. Voice & tone
 - 2. BAC estimation (Widmark approximation)
 - 3. Impairment escalation matrix
 - 4. Disclaimer policy
 - 5. Recovery Mode
 - 6. Orb overlay
 - 7. Test invariants
- [\[\] artifacts/aforce-os/docs/investor-demo-readiness.md](#)
- AForce OS — Investor Demo Readiness
 - Ready now (works today, in this build)
 - Done this sprint (Phase 10)
 - Before TestFlight (internal beta) — owner / Apple actions
 - Before Miami (October field demo)
 - Before public launch (January)
 - Risk level

- Next sprint (suggested, not committed)
- [artifacts/aforce-os/docs/TESTFLIGHT_CHECKLIST.md](#)
- AForce OS — TestFlight Submit Checklist
 - Phase 0 — Prerequisites (do once)
 - Phase 1 — App Store Connect setup (do once)
 - Phase 2 — Pre-build sanity checks
 - Phase 3 — Build with EAS
 - Phase 4 — Submit to TestFlight
 - Phase 5 — Iterate
 - Common gotchas
 - Useful links
- [docs/competitive-moat.md](#)
- AForce — Competitive Moat & Defensibility
 - 1 · The Structural Thesis
 - 2 · Competitor-by-Competitor Defense
 - 3 · Surface-by-Surface Defense Map
 - 4 · The Asymmetry, In One Sentence Per Layer
 - 5 · What Would It Take For Each Player to Catch Up?
 - 6 · The One-Line Closer
- [docs/validation-methodology.md](#)
- AForce OS — Validation Methodology
 - 1. Sweat rate (per session)
 - 2. Sodium replacement bands
 - 3. Heat Index (Heat Guard activation)
 - 4. Estimated blood alcohol concentration (Social Mode)
 - 5. Recovery & readiness adjustments (Apple Health overlay)
 - 6. Per-event hydration impact (the score itself)
 - 7. Compliance streak & retention
 - How to validate this in the field
- [artifacts/api-server/docs/scaling-architecture.md](#)
- AForce OS — Scaling Architecture (50M+ Concurrent Users)
 - 1. Design principles
 - 2. Layer map
 - 3. Services (boundaries + scaling pattern)
 - 4. Database strategy
 - 5. Caching rules
 - 6. Event-driven core
 - 7. Real-time delivery
 - 8. AI decisioning at scale
 - 9. Rate limiting + abuse protection
 - 10. Observability
 - 11. Failover strategy
 - 12. Load testing
 - 13. Performance targets (SLOs)
 - 14. Codebase conventions
 - 15. Deployment
- [artifacts/api-server/docs/load-testing-plan.md](#)
- AForce OS — Load Testing Plan
 - Objective
 - Tooling
 - Stages
 - Scenarios
 - Bottleneck detection process
 - Acceptance gates
 - Schedule
- [artifacts/api-server/docs/failover-strategy.md](#)

- AForce OS — Failover & Resilience Strategy
 - Failure domains
 - Multi-region topology
 - Circuit breakers
 - Retries with backoff
 - Degraded modes
 - Kill switches
 - Traffic shedding
 - Runbooks
 - Recovery objectives
 - Game days
- [artifacts/api-server/loadtests/spec.md](#)
- Load Test Spec

replit.md

Overview

Brand system: v2.1.0 (AForce Brand System) — see [design/aforce-design-tokens.md](#) .

AForce OS is a real-time human performance operating system, delivered as a React Native / Expo mobile application with an Express 5 and PostgreSQL API server. Its core purpose is to provide hydration intelligence and AI-driven insights to enhance athletic performance and overall wellness. Key features include personalized hydration tracking, AI coaching, social engagement through "Circles" and "Territory" features, and integrated e-commerce capabilities via Stripe for purchases and subscriptions. The system is designed for horizontal scaling.

FINAL BUILD LOCK (locked; do not redesign)

Approved for implementation. No redesign, no rebuilding, no moving navigation, no adding tabs, no dashboard expansion. Build once. Expose over time.

AUTHORIZED OVERRIDE — Home + Navigation redesign (owner-approved)

The owner explicitly authorized a full redesign of the **Home screen and tab navigation**, overriding the "do not redesign / do not move navigation / no dashboard expansion" clauses above *for these two surfaces only*. The rest of the lock (Water-First, Score Protection, Language lock, Product Positioning) remains in force and the redesign was built to comply with it.

- **Accent:** added `accent.brand #FF3B30 (+ brandGlow / brandDim / brandSubtle )` used sparsely (thin lines, eyebrows, progress fills). `tabBar.active` flipped to brand red. `accent.primary` and all WHOOP recovery/state colors are untouched, so DEPLETED red still reads.
- **Navigation 7 → 5 visible tabs:** Home · Hydration (was Journal, relabeled + droplet icon) · Protocols · Community · Profile. `scan / social / social-legacy / sleep` are hidden (`href:null` in `ClassicTabLayout`, omitted from `NativeTabLayout` triggers) — engines stay, surfaces hidden, consistent with "Build 100% · Show 10%".
- **Home expansion:** the score orb is retained as the **Readiness Score** hero (thin `READINESS SCORE` eyebrow; engine machinery unchanged). Below it, the Hydration Status card now occupies the **Water Cycle** slot. (Owner trim, June 2026: the Daily Ritual rail, Today's Protocol, Streak, Athlete Mode progress, and Membership cards were removed from the Home surface. A later owner change, June 2026: the WATER CYCLE 8-cell telemetry bar was removed from Home and the Hydration Status card was moved into its place. All removed components — the five cards, the `WaterCycleBar` (still imported by the dormant `SignalsZone`), and the `utils/homeDashboard` derivations — remain in the codebase, simply no longer rendered on Home, so the trim is reversible.)
- **Score-Protection preserved:** every Home surface is a read-only projection of already-completed behaviour via the pure `utils/homeDashboard.ts` helpers (unit-tested); nothing awards or inflates score. (Owner trim, June 2026: the Ritual/protocol rails and the "SOON" Points/Challenges/Referrals preview tiles were removed from the Home surface along with the cards noted above. The retained Hydration Status ring stays a read-only projection of logged intake; the trimmed derivations remain in the codebase so the trim is reversible.)

Cold-Launch Opening Sequence (owner-approved)

A full-screen 4-stage cinematic (`components/opening/OpeningSequence.tsx`) plays once **per cold launch** as an overlay — mounted in `app/_layout.tsx` (AppShell), mirroring the InvestorDemo overlay pattern, so it touches **no routing** (no redirect-loop risk). It is the first of **two stacked cold-launch overlays** driven by the AppShell phase machine (`opening → welcome → done`): when the cinematic finishes it crossfades into the `WelcomeHero` photo front door, and only after the user picks an entry do both overlays dismiss to reveal whatever the app routed to underneath.

- **Stages:** (1) white water-drop symbol w/ breathing fade → (2) AFORCE wordmark + brand-red hairline + "Performance Is Non-Negotiable" → (3) PAUSE/HYDRATE/LOCK IN/PERFORM ritual stagger → (4) "TODAY'S READINESS" + count-up to the live score + a **band-aware** caption via `readinessLabel(performanceState.level)` (a DEPLETED user reads "REHYDRATE NOW", never "READY TO PERFORM").
- **Motion:** slow Apple-Vision-Pro pacing, crossfading absolute-fill layers, no bounce. Tap-anywhere-to-skip; fully reduced-motion aware (`AccessibilityInfo`).
- **Score-Protection:** display-only projection of `engine.score` (DEFAULT_SCORE=92 fallback before state loads); never awards, mutates, or fabricates score.
- **Front door (two overlays, retained):** the `WelcomeHero` photo front door (`components/welcome/WelcomeHero.tsx`) is an intended, retained surface — it is **not** removed and the cinematic is **not** the sole opening surface. It is mounted in `_layout.tsx` *under* the cinematic (zIndex 999 vs 1000) so the cinematic's fade-out crossfades into it with no black flash. When AppShell advances `opening → welcome` the front door's staggered entrance fires; its **GET STARTED** routes to `/onboarding` and **SIGN IN** to `/auth/sign-in`, which advances the phase to `done` and dismisses both overlays to reveal the routed app. Onboarding stays intact — new users get `opening → welcome → onboarding`.
- **Plays per JS launch:** `OpeningMount` holds `useState(true)` initialised once when AppShell mounts; `onFinish` is a stable `useCallback` and `OpeningSequence` keeps it in a ref so the engine-score refresh (on mount + ~30s) never tears down the timeline mid-play.

Water-First Command System

Recommendation order is **Water → Command → Optional support → Score Update**.

Products never come before water. Default recommendation copy must begin with `HYDRATE NOW` / `Start with water` ; optional hydration support may be suggested only after hydration needs are evaluated. Behavior first, product second.

Score Protection Rule

Only completed actions modify score. Recommendations, scans (HydroScan stays advisory), and product selection never increase score. Scores only change from completed behavior.

Language / Localization Lock

Launch languages: English, Spanish, French, German, Portuguese, Italian. No country-specific prioritization. Architecture stays modular so future languages can be added without rebuild. Hidden locales remain resource-only behind flags; they are not in the LanguageSelector.

Engine / UI Governance

Architecture may expand. Navigation may not. The engine becomes smarter; screens remain simpler. Feature flags control exposure. Internal preview stays available. Public unlocks ship phase-by-phase.

Product Positioning Rule

Decision order: Context → Recovery → Behavior → Learning → Optional support. Products support behavior; products do not drive behavior. Never force product recommendations.

MVP Surfaces (do not remove)

Orb · Timeline · HydroScan · Coach · Journal · Recovery · Feature Flags · Internal Preview.
Principle: Build 100% · Show 10% · Unlock over time. One Engine. Multiple Experiences.

Mantra: **Pause → Hydrate → Lock In → Perform.**

User Preferences

I prefer iterative development, with frequent, small updates. Ask before making major changes.

System Architecture

Core Technologies

- **Monorepo:** pnpm workspaces.
- **Backend:** Node.js (v24), Express 5, PostgreSQL, Drizzle ORM, Zod.
- **Mobile:** React Native / Expo SDK 54 with Expo Router 6, React Native Reanimated 3, `i18next` , `@tanstack/react-query` .
- **State Management (mobile):** React Context + `useReducer` (slice-based store).
- **API Tools:** Orval for OpenAPI codegen, generating React Query hooks.

UI/UX Decisions

- **Color Scheme:** AForce Brand System (v2.1.0) — near-black `#0D0D0D` canvas, Signal Red `#C1281B` hero accent used sparingly (thin lines, eyebrows, active states, CTAs), Soursop green `#1FA35A` for positive status (Peak / Optimal / success), Berry blue `#1E5BFF` for secondary data, near-invisible borders, and the four performance-state colors (green / teal / amber / red). Pure black `#000000` is reserved for scrim, drop-shadow color, and `text.inverse` (text on light/accent fills) only. Design tokens exported via `design/aforce-tokens.json` (Tokens Studio format) and human-readable `design/aforce-design-tokens.md`.
- **Design Language:** AForce cinematic dark aesthetic. Content floats on near-black `#0D0D0D`. Data-forward: big numbers, small tracked labels. Soft radial glows, never hard box shadows. Generous spacing throughout. Three type faces by role: Archivo Black (display), IBM Plex Mono (eyebrows / metrics), Inter (body).
- **Visuals:** Stylized maps for "Territory" and smooth animations with Reanimated.

Authentication & Identity

- **Provider:** Clerk (`@clerk/expo` for mobile, `@clerk/express` for server).
- **Sign-in:** Custom email/password and Google SSO.
- **Integration:** Token bridging between Clerk and the OpenAPI client.
- **Auth-gated routes:** Implemented on both mobile and server-side.

Subscription & Entitlement

- **Source of Truth:** Stripe, with webhook events mirrored to PostgreSQL via `stripe-replit-sync`.
- **Client-side:** `useEntitlement.ts` hook for paywall gating based on plan tier.
- **Security:** Server-side computation for pricing, shipping, and tax; webhook signature verification.

Publishing & Distribution

- **Build service:** Expo Application Services (EAS Build) — configured in `artifacts/aforce-os/eas.json` with `development`, `preview`, and `production` profiles.
- **Bundle IDs:** iOS `com.aforce.os`, Android `com.aforce.os` (set in `app.json`).
- **App icon / splash / adaptive icon:** branded set in `assets/images/` (`icon.png`, `splash.png`, `adaptive-icon.png`, `favicon.png`). All four render the **N-N "Non-Negotiable" monogram** — two heavy geometric N's facing each other (left forward, right mirrored) in Bone `#F5F0E8` with a Signal Red `#C1281B` center bar, on Cinematic Black `#0D0D0D`. `icon.png` (1024²) has a subtle charcoal `#1A1A1A` → black radial vignette; `adaptive-icon.png` (1024²) keeps the mark inside the Android safe zone on solid black; `splash.png` (1242×2436) centers the mark on solid black; `favicon.png` (196²) matches. No green, no gradients on the letterforms, no "AForce OS" text in the mark. The marks are generated as a font-free vector SVG rendered via `rsvg-convert` (regenerate from vector — do not AI-generate or the geometry/colors drift). `app.json` paths + `#0D0D0D` backgrounds are already wired.
- **Build commands** (run from `artifacts/aforce-os/` after `pnpm eas:login` + `pnpm eas:configure`):
 - iOS production: `pnpm eas:build:ios`
 - Android production: `pnpm eas:build:android`
 - Both: `pnpm eas:build:all`
 - Internal preview (no store submission): `pnpm eas:build:preview`
- **Submit commands:**
 - iOS App Store: `pnpm eas:submit:ios`
 - Google Play (internal track): `pnpm eas:submit:android`
- **One-time setup needed before first submit:**
 - **iOS** — instead of hand-editing `eas.json`, run the helper from repo root:

```
EAS_ASC_APP_ID=1234567890 \  
EAS_APPLE_TEAM_ID=ABCDE12345 \  
pnpm --filter @workspace/scripts run eas-configure-submit
```

The script validates formats (ASC ID = numeric, Team ID = 10 uppercase alphanumeric), is idempotent, and refuses placeholders. Find your two IDs at: App Store Connect → My Apps → AForce OS → App Information (ASC App ID) and developer.apple.com → Membership (Team ID).

- **Android** — place a Google Play service account JSON at `artifacts/aforce-os/google-service-account.json` (path already in `eas.json`); never commit it.

Release-Readiness Status (verified)

- **Stripe webhook** — wired end-to-end via the Replit Stripe connector. The webhook secret is pulled from `settings.webhook_secret` on the connector (NOT from a `STRIPE_WEBHOOK_SECRET` env var). Verified by server boot log: `initStripe: managed webhook ensured` + `initStripe: syncBackfill complete`. No env var to set.
- **EAS config** — `eas.json` profiles (development/preview/production) and submit blocks are complete; only the two iOS submit IDs need filling via the helper script above. Bundle IDs (`com.aforce.os`), permissions, splash, and icon are all set in `app.json`.
- **App Store / Play Store screenshots** — Apple and Google only accept screenshots captured from a real device or simulator running a built binary. Replit's web preview of the Expo app cannot produce submission-grade assets. Capture path:
 1. `pnpm --filter @workspace/aforce-os run eas:build:dev` to produce an iOS Simulator build.
 2. Boot the build in Xcode Simulator on the required device families (6.7" iPhone, 6.5" iPhone, 13" iPad).
 3. Capture with `xcrun simctl io booted screenshot <name>.png`.
 4. For Android, run `eas:build:preview` to get an APK, install on a Pixel emulator, capture via Android Studio.

Server Hardening

- **Routing:** Express 5 with Zod input/output validation based on OpenAPI spec.
- **Concurrency:** `SELECT ... FOR UPDATE` for serializing concurrent user actions.
- **Rate Limiting & Cache:** OpenWeather API proxied with in-memory TTL cache; rate limits on public and authenticated endpoints.
- **Logging:** Structured logging for request handlers and non-request code.
- **Real-time:** REST mutations are broadcast over a shared HTTP/WebSocket server.

Technical Implementations & Feature Specifications

- **Persistence & Real-Time Backend:** PostgreSQL via Drizzle ORM; REST API broadcasts mutations to WebSocket clients.
- **Per-Event Hydration Scoring:** Defined point values, absorption caps, and release curves.
- **AForce Protocol Screen:** Synchronous derivation of protocol stage based on user state.
- **Water Cycle / "Become AForce":** Modals for tracking water and hydration stick intake, with flavor inference.
- **Social Mode:** Alcohol intake impacts hydration scores and decay rates.
- **Multi-Provider Health Signals:** Integration with various health platforms (e.g., Apple Health, Oura) using a "freshest-wins" logic.
- **Hydration Depletion Math:** Pure, dependency-free helper for score-points-per-minute decay based on physiological standards.
- **AI Coach Voice Engine:** Utilizes ElevenLabs for voice output, providing verdict-aware comparisons after HydroScans and guiding users through performance events. Features include status color systems, video overlay voices, and a "Cinematic v2" refinement with a playback lifecycle state machine.

- **Investor Demo:** A scripted 60-second full-screen overlay showcasing the Voice Engine's states and capabilities for investors.
- **API Server:** Designed for horizontal scaling, includes Stripe integration, auth-gated routes, and social graph routes.
- **Store + Subscription System:** Defines SKU pricing, discounts, bundles, and five consumer subscription tiers with feature gating.

Architecture Diagram

- **app/** : Root layouts, screens, tab bar, gated routes.
- **components/** : Reusable UI elements.
- **services/** : Business logic.
- **store/** : Slice-based reducer state.
- **utils/** : Pure helpers.
- **featureFlags/** : Feature toggles.
- **theme/** : AForce Brand System (v2.1.0) color system, typography, spacing, radii, shadows, status color engine.
- **design/** : Figma design tokens (`aforce-tokens.json`) and human-readable spec (`aforce-design-tokens.md`). Download endpoint: `GET /api/design-tokens` . Design guide: `GET /api/design-guide` .
- **types/** : Global type definitions.
- **data/** : Mock data, product definitions, templates.

External Dependencies

- **Stripe:** Payment processing, subscriptions.
- **stripe-replit-sync:** Stripe webhook mirroring to PostgreSQL.
- **Clerk:** Authentication service.
- **Expo SDK 54:** React Native development framework.
- **Expo WebBrowser / AuthSession:** OAuth, in-app browser.
- **Expo Speech:** Text-to-speech fallback.
- **@expo-google-fonts/inter:** Custom font.
- **React Native Reanimated:** Declarative animations.
- **React Native Gesture Handler:** Gesture recognition.
- **PostgreSQL:** Primary database.
- **Drizzle ORM:** Schema and query layer.
- **Orval:** OpenAPI codegen.
- **Zod:** Schema validation.
- **pnpm workspaces:** Monorepo management.
- **esbuild:** Bundling.
- **OpenWeather API:** Environmental data.
- **ElevenLabs:** Text-to-speech service.
- **i18next:** Localization.

📄 artifacts/aforce-os/docs/AForce-OS-Specification.md

AForce OS — Product & Technical Specification

Current as of June 23, 2026. Brand system v2.1.0. This document is compiled from the live codebase (`artifacts/aforce-os` , `artifacts/api-server` , `lib/`) and the locked product decisions in `replit.md` . Where the code and older spec PDFs disagree, the code is treated as the source of truth.

1. Product Overview

AForce OS is a real-time human-performance operating system delivered as a React Native / Expo mobile app backed by an Express 5 + PostgreSQL API server. Its core job is **hydration intelligence**: it turns logged fluid intake, biometrics, and environment into a single live performance score, then coaches the user with an AI voice engine. Around that core it layers recovery, social accountability, competition, and an integrated store/subscription business.

Positioning mantra: *Pause → Hydrate → Lock In → Perform*. **Tagline:** *Performance Is Non-Negotiable*.

2. Product Locks (non-negotiable design rules)

These constrain every feature and cannot be silently overridden.

- **Water-First Command System.** Recommendation order is **Water → Command → Optional support → Score update**. Default coach copy must begin with `HYDRATE NOW` / `Start with water` . Products never lead; behavior first, product second.
- **Score Protection.** Only *completed* actions change the score. Recommendations, scans (HydroScan is advisory), and product selection never increase score. Every Home read-out is a read-only projection of already-completed behavior.
- **Language / Localization Lock.** Launch languages: English, Spanish, French, German, Portuguese, Italian. No country-specific prioritization; architecture stays modular so hidden locales can ship later behind flags.
- **Engine / UI Governance.** The engine may grow smarter; navigation may not grow. Feature flags control exposure. Principle: **Build 100% • Show 10% • Unlock over time**. One engine, multiple experiences.
- **Product Positioning.** Decision order: Context → Recovery → Behavior → Learning → Optional support. Products support behavior; they never drive it.

3. Technical Architecture

Layer	Technology
Monorepo	pnpm workspaces
Mobile	React Native / Expo SDK 54, Expo Router 6, Reanimated 3, Gesture Handler
Mobile state	React Context + <code>useReducer</code> (slice-based store)

Layer	Technology
Data fetching	@tanstack/react-query with Orval-generated hooks
Backend	Node.js 24, Express 5, PostgreSQL, Drizzle ORM, Zod
API contract	OpenAPI-first; Orval codegen → React Query hooks + Zod schemas
Real-time	Shared HTTP + WebSocket server; REST mutations broadcast to clients
i18n	i18next
Auth	Clerk (@clerk/expo mobile, @clerk/express server)
Payments	Stripe + stripe-replit-sync (webhooks mirrored to Postgres)
Voice	ElevenLabs TTS (Expo Speech fallback)
Environment	OpenWeather API (server-proxied, cached)

Folder model (mobile): app/ (routes/layouts), components/, services/ (business logic), store/ (reducer slices), utils/ (pure helpers), featureFlags/, theme/ (brand system), design/ (Figma tokens), data/ (SKUs, plans, templates), types/ .

4. App Surfaces — Screens & Navigation

Navigation is **locked to 5 visible tabs**; additional surfaces exist as stacked or hidden routes (engines built, surfaces revealed over time).

Visible tabs: Home · Hydration (Journal) · Protocols · Community · Profile.

All routes (app/):

- Tabs: index (Home), protocol , scan (HydroScan), journal , competition , social , sleep , profile , social-legacy (*several hidden via href:null*).
- Auth: (auth)/sign-in , (auth)/sign-up (Clerk).
- Hidden / staged: (hidden)/cruise/* (journey · port · excursion · recovery), circles/[id] · manage · shared , heat/guardian , ring/session , subscription/manage , legal/privacy · terms · health-disclaimer , clutch (team command grid), phantom (wearable status), territory (performance map), urine-check , onboarding , weekly-report .

Home composition (current, post-revert orb-focused layout): readiness eyebrow → status headline → Status Pulse Orb → status label → live status line → consequence line → status-based primary CTA → quick-action grid → Hydration Status card → flag-gated zones (render nothing when off) → pinned "AFORCE" brand footer.

5. Feature Flags & Phased Exposure

Phase 1 (Core) is on by default; Phase 2/3 sub-features ship **off** and light up over time. Verified defaults from featureFlags/flags.ts :

On by default (true): city_competition_enabled , state_competition_enabled , team_competition_enabled , global_leaderboard_enabled , cruise_mode_enabled , sleep_mode_enabled , plus spec flags spec_activation , spec_social , spec_coachV2 , spec_recoveryCircle , spec_notifications , spec_orb , spec_timelineLock , spec_hydroJournal , spec_hydroScan , spec_profileSource , spec_sharedContextLayer , spec_uiFreeze .

Off by default (false): clutch_access_enabled (+ heat/inventory/clip), guardian_intelligence_enabled (+ body_map/alerts), phantom_wearable_enabled , ring_enabled , kids_world_enabled , demo_mode_enabled , metabolic_readiness_enabled (+ glucose), performance_age_enabled , voice_checkin_enabled ,

`intent_capture_enabled` , `performance_statements_enabled` , `offline_intake_outbox_enabled` , all `cruise_*` sub-features, `voice_status_module_visible` , and gated locales `spec_language_ar/zh/ja/ko/hi` .

6. Hydration Scoring Engine

A single 0–100 score drives every coach surface in lock-step.

- **Event-driven scoring.** Each logged intake event has defined point values, **absorption caps** (per rolling time window), and **time-release curves** so a large gulp is absorbed gradually rather than all at once.
- **Continuous depletion.** A pure, dependency-free helper computes score-points-per-minute decay from physiological standards, so the score falls between logs.
- **Daily water target** is derived per user from body weight, activity, and climate (heat/humidity from the weather proxy), not a fixed number.
- **Score Protection** is enforced at the engine boundary: scans and recommendations are advisory and never mutate score; only completed intake/recovery behavior does. Scoring runs **on-device**; the server persists client-submitted snapshots rather than acting as the authority.
- Modifiers: Social/Night Mode (alcohol) raises decay; Heat Guard and low-score states apply small product-specific recovery bonuses.

(Exact point/cap constants live in `utils/scoringEngine.ts` , `services/sweatRateEngine.ts` , and the depletion helpers — parameterized in code, intentionally not frozen here.)

7. Performance-State Color System

Five score bands (single source of truth: `theme/statusColor.ts` , brand v2.1.0). Color is a **signal only** — borders, dots, glows, accents, CTA tint — never a fill.

Band	Range	Calm hex	Meaning
OPTIMAL	85–100	<code>#1FA35A</code> (Soursop green)	Peak — soft wide glow
STABLE	70–84	<code>#3DBE7A</code> (green, lighter)	Holding — subtle glow
DECLINING	50–69	<code>#FFDE00</code> (amber)	Slipping — minimal glow
RISK	30–49	<code>#FF8C1A</code> (orange)	Act now — medium glow
CRITICAL	0–29	<code>#FF2800</code> (signal red)	Depleted — tight intense glow

A **Pressure Mode** variant deepens saturation, raises glow alpha, and speeds the pulse for the same band.

This five-band ladder (`theme/statusColor.ts`) drives the **AI Coach status-color layer** — dots, borders, glows, CTA tint — and the score read-out. A **separate, intentional** four-band *Performance State* ladder (`utils/scoringEngine.ts` → `resolveState` : `PEAK ≥90` / `BALANCED ≥75` / `RECOVERING ≥60` / `DEPLETED`, with its own colors in `theme/colors.ts` `states`) drives the **orb** (pulse / flare-on-peak / collapse-on-depletion), `riskTimer` , and command selection. The two ladders use different mid-band thresholds by design but share the same top green (`#1FA35A`) and bottom red (`#FF2800`); see `design/aforce-design-tokens.md` for both tables.

8. Subscription Tiers

Verified monthly prices from `data/subscriptionPlans.ts` .

Consumer (5 tiers):

Plan	Price	Unlocks
Core	\$9.99/mo	Hydration Control Center, Status Pulse + score, basic protocol, basic AI, logging, smart reminders
Recovery+	\$9.99/mo	Adds post-session Recovery Mode
AForce Athlete	\$19.99/mo	Enhanced AI, personalized protocol, advanced recovery, 90-day trends, competition + city/state/team leaderboards, premium notifications, Metabolic Readiness
Performance Bundle	\$59.99/mo	Flagship: Athlete tier + monthly product drop (1 canister or 2 stick packs) at member pricing, priority AI/insights
AForce Elite	\$99/mo	Guardian Mode (individual), premium analytics, full monthly product bundle, early access, concierge support

Team (Core): Team Core Starter \$49/mo (≤ 25 seats) · Team Core Growth \$99/mo (≤ 50) · Team Core Pro \$149/mo (≤ 100) — roster-aware, group reporting, admin console.

Enterprise — Clutch (live team command): Clutch Starter \$1,000/mo · Clutch Pro \$2,500/mo · Clutch Elite \$5,000/mo.

Enterprise — Guardian (injury-risk protection): Guardian Core \$5,000/mo + \$7,500 setup (6-mo min) · Guardian Elite \$8,000/mo + \$12,500 setup (12-mo min).

Entitlement is gated client-side via `useEntitlement.ts` ; Stripe is the source of truth, mirrored to Postgres.

9. Product Store (Physical SKUs)

Catalog model: one SKU per (format × flavor), plus per-format bundles (`data/pricing.ts` , `data/products.ts`). Subscriber/member pricing is lower than list. Pricing/shipping/tax are computed server-side.

- **Formats:** Stick packs (12 ct) · RTD cans (12 pk) · Canisters (30 servings).
- **Flavors:** Berry · Watermelon · Soursop.
- **Bundles:** flavor-agnostic multi-packs (flavor split chosen at checkout).
- Per-serving sodium and serving sizes are encoded per SKU for the sweat-rate math.

10. Data Model (PostgreSQL / Drizzle)

Table	Key columns
<code>aforce_user_state</code>	<code>user_id</code> (PK), <code>units_consumed_today</code> , <code>oz_target</code> , <code>symptom_state</code> , <code>biometrics</code> (JSONB), <code>social_mode</code> (JSONB), <code>intake_events</code> (JSONB)
<code>aforce_intake_logs</code>	<code>id</code> , <code>user_id</code> , <code>fluid_type</code> , <code>oz_amount</code> , <code>score_before</code> , <code>score_after</code>
<code>aforce_score_snapshots</code>	<code>id</code> , <code>user_id</code> , <code>score</code> , <code>level</code> , <code>recovery_score</code> , <code>captured_at</code>
<code>aforce_users</code>	<code>id</code> , <code>stripe_customer_id</code> , <code>plan_id</code> , <code>subscription_status</code> , <code>referral_code</code>
<code>aforce_circle_users</code>	<code>owner_user_id</code> , <code>member_user_id</code> , <code>status</code>
<code>aforce_analytics_events</code>	<code>event_id</code> , <code>analytics_id</code> , <code>event_type</code> , <code>payload</code>

11. API Surface (api-server)

Express 5 with Zod input/output validation generated from the OpenAPI spec.

Method · Path	Purpose
GET /api/aforce/state	Full UserState snapshot
POST /api/aforce/intake	Log fluid consumption
POST /api/aforce/signals	Update active symptoms (mild/moderate/severe)
GET /api/aforce/journal/timeline	Interleaved snapshots + intake history
POST /api/aforce/social/activate	Start a Social-Mode drinking session
POST /api/aforce/social/cruise	Start 24h Enterprise Cruise Mode
GET /api/aforce/weather	OpenWeather proxy (tempC, humidity) by lat/lon
POST /api/voice/tts	ElevenLabs text-to-speech proxy
POST /api/checkout/session	Stripe Checkout for subscriptions
POST /api/stripe/webhook	Stripe events → updates <code>plan_id</code>

Hardening: `SELECT ... FOR UPDATE` to serialize concurrent user actions; rate limiting on public + authenticated endpoints; in-memory TTL cache on the weather proxy; structured request/non-request logging.

12. Integrations

- **Auth — Clerk.** Custom email/password + Google SSO. JWT via `Authorization: Bearer` (REST) and `?token=` (WebSocket). Token bridging into the OpenAPI client; auth-gated routes on both mobile and server.
- **Real-time — WebSocket** mounted on the shared server (`/api/aforce/ws`) with periodic heartbeats; REST mutations are broadcast to connected clients.
- **Payments — Stripe** via `stripe-replit-sync` ; the webhook secret is pulled from the managed Replit Stripe connector (not an env var). Webhook events mirror subscription state to Postgres.
- **Environment — OpenWeather**, server-proxied with a ~10-minute TTL cache (`OPENWEATHER_API_KEY`).
- **Voice — ElevenLabs** (`eleven_turbo_v2_5`); static phrases cached. Expo Speech is the fallback.

13. Localization

Launch set (6, in the selector): English · Spanish · French · German · Portuguese · Italian. Additional locales (Arabic, Chinese, Japanese, Korean, Hindi) are wired as resources but **gated behind flags** and excluded from the language selector. Architecture is modular so new languages add without a rebuild. (Number/date localization is a known gap — formatting is currently device-locale bound.)

14. Publishing & Distribution

- **Build service:** EAS Build — `development` , `preview` , `production` profiles in `eas.json` .
- **Bundle IDs:** iOS `com.aforce.os` , Android `com.aforce.os` .

- **Branding assets:** icon / splash / adaptive / favicon all render the **N-N "Non-Negotiable" monogram** (Bone #F5F0E8 N's, Signal Red #C1281B center bar, Cinematic Black #0D0D0D), generated from a font-free vector SVG.
- **Build:** `pnpm eas:build:ios` · `:android` · `:all` · `:preview` (from `artifacts/aforce-os/`).
- **Submit:** `pnpm eas:submit:ios` (needs ASC App ID + Apple Team ID via the `eas-configure-submit` helper) · `pnpm eas:submit:android` (needs a Google Play service-account JSON, never committed).
- **Store screenshots** must be captured from a real device/simulator build — the web preview can't produce submission-grade assets.

15. Design System (Brand v2.1.0)

- **Canvas:** near-black #0D0D0D . Content floats; soft radial glows, never hard box shadows. Generous spacing.
- **Accents:** Signal Red #C1281B used sparingly (thin lines, eyebrows, active states, CTAs); Soursop green #1FA35A for positive status; Berry blue #1E5BFF for secondary data. Pure black #000000 reserved for scrims/shadows/inverse text.
- **Type (three roles):** Archivo Black (display) · IBM Plex Mono (eyebrows / metrics) · Inter (body).
- **Tokens:** exported as `design/aforce-tokens.json` (Tokens Studio) + human-readable `design/aforce-design-tokens.md` . Served at `GET /api/design-tokens` and `GET /api/design-guide` .

Source of truth: the AForce OS codebase. Locked product decisions are maintained in `replit.md` ; this spec mirrors them and adds the concrete, code-verified inventory.

artifacts/aforce-os/SPEC-SHEET.md

AForce OS — Product & Technical Specification

A complete specification of the **AForce OS** mobile application — a real-time human-performance operating system delivered as a React Native / Expo app with an Express 5 + PostgreSQL backend. Its purpose: turn hydration into measurable readiness, recovery, and a daily ritual — *Pause → Hydrate → Lock In → Perform*.

At a Glance

App name	AForce OS
Version	1.0.0 (iOS build 1 / Android versionCode 1)
Platform	iOS + Android (React Native / Expo), portrait, dark UI
Bundle ID	<code>com.aforce.os</code> (iOS & Android)

Frontend	Expo SDK ~54, React Native 0.81.5, Expo Router 6
Backend	Node.js, Express 5, PostgreSQL, Drizzle ORM, Zod
Auth	Clerk (@clerk/expo)
Payments	Stripe + stripe-replit-sync
Languages	6 live (EN/ES/FR/DE/PT/IT) + 5 wired-but-hidden
Visible tabs	Home · Hydration · Protocols · Community · Profile

1. Platform & Technology Stack

- **Framework:** Expo SDK ~54.0.27 , React Native 0.81.5 (New Architecture enabled, React Compiler enabled, typed routes).
- **Navigation:** Expo Router ~6.0.17 (file-based; both classic Tabs and native NativeTabs).
- **State management:** Zustand store organized into reducer-style slices (store/useAppStore.tsx , store/appStoreReducer.ts , store/slices.tsx), plus a dedicated cart store (store/useCartStore.tsx).
- **Data fetching:** @tanstack/react-query , consuming a generated OpenAPI client (@workspace/api-client-react).
- **Animation:** React Native Reanimated ~4.1.1 + Worklets.
- **Internationalization:** i18next + react-i18next + expo-localization .
- **Hardware / device:** expo-camera (HydroScan), expo-location (weather), @kingstinct/react-native-healthkit (Apple Health), expo-haptics , expo-audio , expo-notifications , expo-secure-store .
- **Typography assets:** Inter, Bebas Neue, DM Sans (@expo-google-fonts/*).

2. Application Architecture

- **app/** — root layouts, the authenticated tab group, gated routes, modals.
- **components/** — reusable UI (orb, rails, opening sequence, etc.).
- **services/** — business logic (scoring, voice, health, social, i18n).
- **store/** — slice-based Zustand state + reducer.
- **utils/** — pure, dependency-free helpers (scoring math, units, dashboards).
- **featureFlags/** — feature toggles controlling surface exposure.
- **theme/** — AForce Brand System (v2.1.0) color system, typography, spacing, radii.
- **data/** — products, subscription plans, templates, mock data.
- **types/** — shared TypeScript definitions.
- **locales/** — translation resources.

Governance principle: *Build 100% · Show 10% · Unlock over time.* The engine expands; navigation stays fixed. Feature flags gate exposure.

3. Navigation & Screen Inventory

Visible tab bar (5 tabs)

1. **Home** — readiness + hydration command dashboard (app/(tabs)/index.tsx).
2. **Hydration** — chronological hydration/recovery feed (app/(tabs)/journal.tsx).
3. **Protocols** — AForce Protocol guidance (app/(tabs)/protocol.tsx).
4. **Community** — rankings, challenges, battles, teams, map (app/(tabs)/competition.tsx).
5. **Profile** — profile + settings (app/(tabs)/profile.tsx).

Hidden routes (`href: null` — built, not surfaced)

`scan` (HydroScan), `social` (Social Mode), `sleep` (Sleep Mode), `social-legacy` (developer-only).

Stack / modal routes

`/store` , `/cart` , `/subscription` , `/achievements` , `/onboarding` , `/circles` , `/territory` , `/ring` , `/heat` .

4. Onboarding & Cold-Launch Opening Sequence

- **Opening sequence** (`components/opening/OpeningSequence.tsx`) — a 4-stage cinematic that plays **once per cold launch** as an overlay (touches no routing):
 1. White water-drop symbol, breathing fade.
 2. AForce wordmark + brand-red hairline + "Performance Is Non-Negotiable".
 3. PAUSE / HYDRATE / LOCK IN / PERFORM ritual stagger.
 4. "TODAY'S READINESS" + count-up to the live score, with a band-aware caption (a depleted user reads "REHYDRATE NOW", never "READY TO PERFORM"). Slow Apple-Vision-Pro pacing, tap-to-skip, fully reduced-motion aware. Display-only (never mutates score).
- **First-run onboarding** (`app/onboarding.tsx`) — Goal selection (Performance / Recovery / Endurance / Balance / Longevity) → Activity level (Sedentary → Athlete) → Profile setup (weight / height / sex). Gated by two flags (`hasSeenWelcome` + `hasCompletedOnboarding`).

5. Hydration Scoring Engine

Files: `services/hydrationScoreService.ts` , `utils/hydrationScore.ts` , `utils/scoringEngine.ts` .

- **Base unit:** 1 unit = 12 oz.
- **Point values:**
 - Water: +0.5 pts/oz (+6 per unit).
 - AForce Berry Blast / Watermelon Surge: +10 pts.
 - AForce Soursop Edge: +11 pts.
- **Contextual bonuses:**
 - *Heat Guard:* +2 for Watermelon Surge when heat score ≥ 45 .
 - *Depleted boost:* +2 for Soursop Edge when score < 40 .
 - *Hydration Cycle:* +8 to +20 for logging water + AForce within 30 min (larger bonus at lower scores).
- **Absorption caps:** max 1.5 units per rolling 20-min window; excess absorbed at 75% efficiency.
- **Release curves:**
 - Water: 60% immediate, 40% released over ~12.5 min.
 - AForce: 70% immediate, 30% released over ~25 min.
- **Performance states (4-band, from `utils/scoringEngine.ts` → `resolveState`):** PEAK (90–100), BALANCED (75–89), RECOVERING (60–74), DEPLETED (0–59). This ladder drives the orb (pulse / flare / collapse), `riskTimer` , and command selection. It is **distinct from** the 5-band *Score Status* color ladder (OPTIMAL / STABLE / DECLINING / RISK / CRITICAL, `theme/statusColor.ts`) that tints the AI Coach status surfaces and the score read-out. Hex values for both live in `docs/AForce-OS-Specification.md` §7 and `design/aforce-design-tokens.md` .
- **Score Protection rule:** only *completed* behavior changes the score. Recommendations, scans (HydroScan is advisory), and product selection never increase the score.

6. Recovery & Health Signals

Files: `services/healthConnection.ts` , `services/recoveryEngine.ts` , `utils/biometricsAggregator.ts` .

- **Providers:** Apple Health (HealthKit), Samsung Health (SDK), WHOOP (OAuth), merged "freshest-wins".
- **Recovery Capacity engine:** derives a 0-100 score from hydration score minus penalties (sleep loss, dark urine, drink load) plus a streak boost.
- **Tracked signals:** thirst (1-5), energy (1-5), urine color (1-8), steps, temperature, humidity, workout load.
- **Apple Health entitlements:** reads heart rate, HRV, sleep, and workouts; writes hydration logs back to Health.

7. AI Coach / Voice Engine

Files: `services/elevenLabsTts.ts` , `services/voiceService.ts` , `services/intentClassifier.ts` .

- **Stack:** ElevenLabs TTS proxied through the API server (`/api/voice/tts`), with `expo-speech` as fallback.
- **Lifecycle:** classify transcript intent (e.g., `LOG_INTAKE` , `COMPLETE_CYCLE`) → resolve persona by performance band → render template → TTS output.
- **Persona modes:** tone and phrasing shift with hydration band (more decisive in lower bands); verdict-aware comparisons after HydroScans.
- **Water-First copy lock:** coach recommendations always begin with `HYDRATE NOW` / `Start with water` ; optional product support is only suggested after hydration needs are evaluated.

8. Social Layer

Files: `services/circleService.ts` , `services/recoveryCircle.ts` , `services/territoryEngine.ts` .

- **Circles:** accountability groups for friends and teams.
- **Recovery Circle:** private cohort (max 3 people) with fixed checkpoints (Day 0, 1, 3, 7, 30).
- **Territory:** competitive regional scoring — $\text{Avg Performance} \times 0.35 + \text{Protocol Rate} \times 0.25 + \text{Streak Density} \times 0.15 + \text{Recovery Efficiency} \times 0.15 + \text{Momentum} \times 0.10$.
- **Social Mode:** alcohol intake affects hydration score and decay rate.

9. Store & Subscription System

Files: `data/subscriptionPlans.ts` , `types/subscription.ts` . *Plans inherit features via an inheritance chain; feature flags map to `featureFlags/flags.ts` .*

Consumer

Plan	Price	Notes
Core	\$9.99/mo	Entry: OS access, AI commands, daily tracking.
Recovery+	\$9.99/mo	Standalone add-on — unlocks Recovery Mode after Social Mode.
AForce Athlete	\$19.99/mo	Personalized protocols, advanced recovery, competition access.
Performance Bundle	\$59.99/mo	Best value — Athlete + monthly product drop (1 canister or 2 stick packs).

Plan	Price	Notes
AForce Elite	\$99/mo	Guardian Mode for individuals + full monthly bundle (canister + sticks + RTDs) + concierge.

Team / Program

Plan	Price	Seats
Team Core Starter	\$49/mo	up to 25
Team Core Growth	\$99/mo	up to 50
Team Core Pro	\$149/mo	up to 100

Performance Systems — Clutch Access

Plan	Price
Clutch Starter	\$1,000/mo
Clutch Pro	\$2,500/mo
Clutch Elite	\$5,000/mo

Performance Systems — Guardian

Plan	Price	Terms
Guardian Core	\$5,000/mo	+ \$7,500 setup, 6-month minimum
Guardian Elite	\$8,000/mo	+ \$12,500 setup, 12-month minimum

- **Pricing security:** pricing, shipping, and tax are computed server-side; Stripe is the source of truth, mirrored to PostgreSQL via webhooks. Webhook signatures are verified.
- **Entitlements:** `useEntitlement.ts` gates paywalled surfaces by plan tier.

10. Authentication & Identity

- **Provider:** Clerk — `@clerk/expo` (mobile), `@clerk/express` (server).
- **Sign-in:** custom email/password + Google SSO.
- **Gating:** the `(tabs)` group is gated behind `isSignedIn` ; `ClerkProvider` mounts at the root layout. Auth-gated routes exist on both client and server.
- **Token bridging:** Clerk session tokens bridge into the OpenAPI client.

11. Design System

Files: `theme/colors.ts` , `theme/typography.ts` , `theme/spacing.ts` .

- **Palette (AForce Brand System):** near-black `#0D0D0D` canvas, AForce signal red `#C1281B` (hero accent, used sparingly), Soursop Edge green `#1FA35A` (Peak / positive status), Berry blue `#1E5BFF` (secondary / info), plus the score-band ladder (green → amber → orange → red).
- **Typography:** Archivo Black (display), IBM Plex Mono (eyebrows / metric labels), Inter (400 → 800) body, tracked labels (letter-spacing up to "ultra").
- **Spacing:** 4px grid (scale 1-24; e.g. step 8 = 32px).
- **Aesthetic:** content floats on the near-black canvas, big numbers / small tracked labels, soft

radial glows (no hard box shadows), generous spacing.

- Tokens exported in Tokens Studio format (`design/aforce-tokens.json`) and a human-readable spec (`design/aforce-design-tokens.md`).

12. Localization

- **Live (visible in selector):** English, Spanish, French, German, Portuguese, Italian.
- **Wired but hidden behind flags:** Arabic, Chinese, Japanese, Korean, Hindi.
- **Lock:** no country-specific prioritization; architecture stays modular so new languages add without a rebuild. *Files:* `services/i18nService.ts` , `locales/` .

13. Feature Flags

File: `featureFlags/flags.ts` . *Two profiles:* `DEFAULT_FLAGS` and `DEMO_ALL_ON_FLAGS` . Representative flags: `clutch_access_enabled` , `guardian_intelligence_enabled` , `phantom_wearable_enabled` , `cruise_mode_enabled` , `global_leaderboard_enabled` , `team_roster_enabled` . Flags map to subscription feature gating so entitlements stay consistent.

14. Backend & API

- **Server:** Express 5 with Zod input/output validation derived from an OpenAPI spec (Orval generates the React Query client + Zod schemas).
- **Database:** PostgreSQL via Drizzle ORM.
- **Concurrency:** `SELECT ... FOR UPDATE` to serialize concurrent user actions.
- **Real-time:** REST mutations broadcast over a shared HTTP/WebSocket server.
- **Integrations:** Stripe (+ `stripe-replit-sync`), OpenWeather (proxied with an in-memory TTL cache + rate limiting), ElevenLabs voice (`/api/voice/tts`).
- **Logging:** structured logging for request and non-request code.
- **Scaling:** designed for horizontal scaling.

15. Build, Distribution & Permissions

- **Build service:** EAS Build — `development` , `preview` , `production` profiles (`eas.json`).
- **Bundle IDs:** iOS & Android `com.aforce.os` ; iOS `supportsTablet: false` .
- **Versioning:** app version 1.0.0; iOS buildNumber 1; Android versionCode 1.
- **App config:** dark UI, portrait, dark splash, notification accent `#C1281B` .
- **Permissions:**
 - iOS — Camera (barcode scan), Location When In Use (weather), HealthKit (read HR/HRV/sleep/workouts; write hydration logs).
 - Android — `CAMERA` , `ACCESS_COARSE_LOCATION` , `ACCESS_FINE_LOCATION` .
- **Build commands** (from `artifacts/aforce-os/`): `eas:build:ios` , `eas:build:android` , `eas:build:all` , `eas:build:preview` , `eas:build:dev` ; submit via `eas:submit:ios` / `eas:submit:android` .

Appendix — Key Dependency Versions

Package	Version
expo	~54.0.27
react-native	0.81.5
expo-router	~6.0.17
react-native-reanimated	~4.1.1
@clerk/expo	3.2.2
@kingstinct/react-native-healthkit	^14.0.0
@tanstack/react-query	catalog
i18next	^26.0.6
react-i18next	^17.0.4
expo-camera	~17.0.10
expo-audio	~1.1.1
expo-speech	~14.0.8
lucide-react-native	^1.16.0
typescript	~5.9.2

Specification compiled from the AForce OS codebase. Internal — for discussion and planning purposes.

☐ AFORCE_FINAL_SPEC.md

AFORCE OS — Final Spec (Core Product)

Brand system: v2.1.0 (AForce Brand System) — see `design/aforce-design-tokens.md` .

Authoritative core-product reference. This document captures the baseline AForce OS product. Social Mode and Cruise Mode enhancement layers live in a separate document — `AFORCE_SOCIAL_CRUISE_ADDON.md` — and **must not be merged into this spec** under any circumstance.

Implementation Order

Implement one phase at a time. Stop after each phase. Never rebuild. Analyze current code first. Protect existing architecture.

#	Phase	Status
1	Opening Screen Safe-Area Fix	—
2	Profile + Units + Login	—
3	Bottom Navigation + Timeline	—

#	Phase	Status
4	HydroScan Core	—
5	Orb Intelligence	—
6	Heat + Territory	—
7	Share System + BECOME AForce footer	—
8	Legal + Compliance	—
9	Feature Locks (Guardian / Clutch hidden)	—
10	Investor Demo Overlay	Pending

Live status is tracked in `AForce_PHASE_STATUS.md`.

Hard Rules

- One phase per session. Stop after the phase ships. Wait for approval.
- Never rebuild existing surfaces. Patch, do not redesign.
- Analyze current code first. Touch only files relevant to the phase.
- Protect the existing architecture (pnpm monorepo, Expo Router stack, AForce cinematic dark aesthetic, slice-based store, Drizzle/Zod server contracts).
- Social Mode and Cruise Mode additions stay in the addon document and are not implemented until **all** core phases are stable.

Final Refinement Locks

Refinement locks only — **not** architecture changes. No redesign, no new navigation, no new tabs, no new screens, no dashboard expansion. Goal: increase retention, reduce friction, protect free mode, improve future Phantom rollout. Keep AForce OS simple externally and powerful internally.

Lock #1 — Impact Engine confidence. Impact confidence increases as signal quality improves. Impact uses existing surfaces only — no new UI. *Purpose:* protect the loop — Signal → Command → Action → Impact → Learning. No rebuilds required.

Lock #2 — Manual-first engine rule. The core hydration loop must function using phone + manual inputs only. Wearables and future Phantom signals increase confidence but are not required. *Purpose:* protect free users, protect onboarding, prevent hardware dependency.

Lock #3 — Reminder guardrail. Completed goals reduce correction urgency and reminder frequency. *Purpose:* reduce notification fatigue, improve long-term retention.

Lock #4 — Verification layer rule. The verification layer resolves to the highest-confidence signal source available and degrades gracefully downward. Priority: (1) Phantom Signals → (2) Connected Wearables → (3) Phone + Manual Inputs. *Purpose:* protect the future Phantom rollout without creating dependencies.

These are refinement locks only. No architecture rebuilds. No additional user-facing modules.

Phase 10 — Investor Demo Overlay

Status: Pending

A scripted 60-second full-screen cinematic overlay. Triggered only when the `demo_mode_enabled` feature flag is ON. Never visible in production. Never accessible via standard navigation.

Script — 6 acts, 10 seconds each

Act 1 — Opening (0:00-0:10) Black screen. AForce wordmark fades in — Bone `#F5F0E8` with a Signal Red `#C1281B` hairline on near-black `#0D0D0D`. Subtitle: "The Performance Operating System." Orb pulses once.

Act 2 — Readiness Score (0:10-0:20) Orb animates from Depleted → Recovering → Balanced → Peak. Score climbs from 14 to 97. Soursop-green peak glow (`#1FA35A`) intensifies. Label: "From depleted to peak. In real time."

Act 3 — HydroScan (0:20-0:30) Scan animation. Product recognized. Score updates. Voice Engine fires: "You're back in range. Lock in." Label: "AI-powered hydration intelligence."

Act 4 — Social Mode (0:30-0:40) BAC overlay appears on orb. Crimson ring pulses. Safety card visible. Recovery Mode activates. Label: "Performance protection. Even off the clock."

Act 5 — Territory + Heat Guard (0:40-0:50) Stylized map activates. Heat Guard band flips to WARNING. Guardian alert fires. Label: "Environmental intelligence. Real-time."

Act 6 — The Standard (0:50-1:00) All surfaces collapse back to the clean orb at PEAK. One line fades in: "Built for people who don't get to be off." AForce wordmark. Fade to black.

Hard Rules

- `demo_mode_enabled` must be `false` in all production builds.
- No real user data is used — all demo state is seeded from `data/demoProfile.ts`.
- Overlay dismisses on tap at any point.
- Auto-dismisses at 60 seconds — the modal closes to reveal the app underneath (no return to a `welcome.tsx` screen; the cinematic mounts as an overlay over whatever the app routed to).

Product Surface (high-level)

Mobile — `artifacts/aforce-os` (Expo SDK 54 / Expo Router 6)

- **Opening sequence:** no `app/splash.tsx` / `app/welcome.tsx` routes exist. `app/_layout.tsx` (AppShell) runs a cold-launch front-door state machine (`opening` → `welcome` → `done`) of stacked overlays that touch no routing: the `OpeningSequence` cinematic (`components/opening/OpeningSequence.tsx` , top-most) plays once per cold launch, then crossfades into the `WelcomeHero` photo front door (`components/welcome/WelcomeHero.tsx` — GET STARTED → `/onboarding` , SIGN IN → `/auth/sign-in`); once an entry is picked the overlays dismiss to reveal the routed app underneath. The home dashboard is `app/(tabs)/index.tsx` ; first-run routing is decided by `SplashGate` via the pure `firstRunRoute` helper keyed on `hasCompletedOnboarding` .
- **Tab bar:** `app/(tabs)/` — `index` , `profile` , plus the rest of the primary tabs. Bottom navigation includes the Timeline surface.
- **HydroScan:** `app/scan.tsx` → `screens/HydrationScanScreen.tsx` .
- **Orb:** hydration-score orb surface backed by the live scoring engine. Driven by `services/` , animated with `Reanimated 3`.
- **Heat / Territory:** `app/heat.tsx` , `app/territory.tsx` .
- **Share:** `app/share.tsx` → `screens/SharePreviewScreen.tsx` . The invite + "BECOME AFORCE" footer lives in the Profile invite card (spec #7 referral loop).
- **Legal:** `app/legal/` .
- **Feature locks:** Guardian (heat/guardian) and Clutch surfaces are hidden from production navigation until released.

API server — `artifacts/api-server`

- Express 5, Postgres + Drizzle, Zod input/output validation derived from `lib/api-spec/openapi.yaml` .

- Auth-gated routes via Clerk (`@clerk/express`).
- Stripe entitlements mirrored via the managed `stripe-replit-sync` webhook (no `STRIPE_WEBHOOK_SECRET` env var — pulled from the Replit connector at boot).
- OpenWeather proxy with in-memory TTL cache.
- WebSocket broadcaster shared with the HTTP server.

Shared libs — `lib/`

- `api-spec` (OpenAPI source of truth, regenerates `api-client-react`
 - `api-zod` `ON` `pnpm --filter @workspace/api-spec run codegen`).
- `db` (Drizzle schemas + migrations).

Design Language

AForce cinematic dark aesthetic (AForce Brand System v2.1.0). Near-black canvas (`#0D0D0D`), Signal Red hero accent (`#C1281B`) used sparingly, Soursop green (`#1FA35A`) for positive status, an amber → orange → red declining ladder, Berry blue (`#1E5BFF`) for secondary data, near-invisible borders. Soft radial glows, never hard box shadows. Big numbers, small tracked labels. Generous spacing. Typography: Archivo Black (display), IBM Plex Mono (eyebrows / metrics), Inter (body). Pure black (`#000000`) is reserved for scrim, drop-shadow color, and `text.inverse` only. `design/aforce-design-tokens.md` is the canonical source of truth; tokens are also exported via `design/aforce-tokens.json` (Tokens Studio).

Out of Scope (this document)

- Social Mode contexts, Morning Reset, Moments Engine
- Cruise Mode Voyage Recovery, Recovery Concierge, Cruise contexts
- Recovery Journey, Journey Summary, Phantom — **not built**, kept as architecture-only stubs

See `AFORCE_SOCIAL_CRUISE_ADDON.md` for the enhancement layer that sits on top of this core once it is stable.

📄 AFORCE_PHASE_STATUS.md

AFORCE OS — Phase Status

Live tracker for the implementation order defined in `AFORCE_FINAL_SPEC.md` (core) and `AFORCE_SOCIAL_CRUISE_ADDON.md` (enhancement layer).

Update this file at the end of every phase. One phase at a time. Stop after each phase and wait for approval before continuing.

Status legend: `🔲` pending · `🔲` in progress · `🔲` shipped · `🔲` blocked

Core Phases (AFORCE_FINAL_SPEC.md)

#	Phase	Status	Notes
---	-------	--------	-------

#	Phase	Status	Notes
1	Opening Screen Safe-Area Fix	□	Added <code><StatusBar style="light" /></code> once at root layout so system glyphs (clock/battery/signal) stay visible against the pure-black opening canvas. Existing safe-area inset math on <code>app/splash.tsx</code> + <code>app/welcome.tsx</code> was already robust (<code>Math.max(insets.top + 28, winH * 0.08)</code>) and was left untouched.
2	Profile + Units + Login	□	Audit: Profile (Clerk user binding, sign-out button), Units (weight lbs/kg, temp F/C, volume oz/mL with persistent slice + 137 lines of tests), and Login (sign-in 254 lines, sign-up 375 lines with email/password + Google SSO) were already built. Real gap: <code>app/index.tsx</code> had no <code>isSignedIn</code> check — sign-in screens were unreachable. Surgical fix: added auth gate in <code>app/index.tsx</code> (respecting <code>DEMO_MODE</code> so pitch screenshots keep working) + defensive <code>(auth)/_layout.tsx</code> redirect when already signed-in.
3	Bottom Navigation + Timeline	□	Audit: already fully shipped, later reconciled to the locked 5 visible tabs (Home · Hydration · Protocols · Community · Profile) with dual implementations — <code>NativeTabs</code> on iOS (liquid glass) + classic <code>Tabs</code> on Android/web with custom <code>PlainTabButton</code> (haptic tick, AForce Brand System (v2.1.0) styling, Signal Red active tint, transparent <code>BlurView</code> on iOS, 84px web height). The Hydration tab = <code>JournalScreen</code> ("PERFORMANCE TIMELINE" eyebrow, 7/30/90 range picker, section summaries Recovery/Heat/Hydration/Corrections/Territory/Streaks, Win Moments strip, score-over-time chart with band zones, collapsible day cards from <code>/journal/rollups</code> , Export PDF). The Check/scan, Social, and Sleep surfaces are hidden via <code>href:null</code> (engines built, surfaces hidden — "Build 100% · Show 10%"), not deleted. Route file stays <code>journal.tsx</code> for deep-link stability; the user-facing label is now Hydration (droplet icon) via <code>i18n</code> , and Protocols is likewise a user-facing relabel. Store correctly excluded from bottom nav. Navigation reconciled to labels/visibility only — no navigation code changed.
4	HydroScan Core	□	Audit: already fully shipped. Route <code>app/scan.tsx</code> → <code>HydrationScanScreen</code> (907 lines): scan → recognize → score → recommend → log into live store, with success flash overlay (20% PEAK tint + <code>Haptics.Success</code> + <code>router.back</code> at 800ms per spec §11). Companion: <code>app/urine-check.tsx</code> → <code>UrineHydrationCheckScreen</code> (261 lines). Services: <code>hydrationScanService</code> , <code>hydrationScoreService</code> , <code>hydrationStatus</code> , <code>productRecognitionService</code> , <code>scanCoachVoice</code> (174 lines, voice-coach script builder), <code>urineHydrationCheck</code> , <code>beverageComparisonEngine</code> (204 lines), <code>openFoodFactsService</code> . Components: <code>ScanResultCard</code> , <code>ScanAICoachCard</code> , <code>ProductFitCard</code> , <code>AForceReplacementCard</code> , <code>CameraScanModal</code> (Expo Camera barcode scanner on native), <code>AddDrinkModal</code> , <code>SmartCaptureModal</code> , <code>WhyThisForYouCard</code> , <code>SuperfoodSignalsCard</code> . Mock barcode tray + manual search field for web preview where Expo Camera is unavailable. Tests: <code>hydrationScanRecommendation.test.ts</code> , <code>scanCoachVoice.test.ts</code> , <code>drinkCatalog.test.ts</code> . No code changes required.
5	Orb Intelligence	□	Audit: already fully shipped. <code>StatusPulseOrb</code> (550 lines, <code>Reanimated 3</code>): pulse fully driven by <code>pulseConfig</code> from service layer — 4 <code>waveBehavior</code> modes plus <code>flareOnPeak</code> (rhythmic accent ring at PEAK), <code>collapseOnDepletion</code> (tense inward squeeze at DEPLETED), <code>burstOnIntake</code> (outward shockwave on every successful intake), continuous secondary ripple ring in BALANCED/PEAK. Tappable to open Score Breakdown sheet. Optional <code>socialOverlay</code> (alcohol load ring, crimson on HIGH/CRITICAL impairment) and <code>displayedAccent</code> (locks orb digit color to tweened display score, while pulse motion still reflects true physiological state). Backed by <code>hydrationScoreService</code> (217 lines) + <code>hydrationStatus</code> (125 lines, <code>getHydrationStatus()</code> returns headline/label/consequence/CTA). Mounted on Home tab (<code>app/(tabs)/index.tsx</code> line 238) inside the 5-step layout (headline → orb → label → consequence → CTA). Tests: <code>hydrationStatus.test.ts</code> , <code>statusColor.test.ts</code> . No code changes required.

#	Phase	Status	Notes
6	Heat + Territory	□	Audit: already fully shipped. Routes <code>app/heat.tsx</code> → <code>HeatRiskScreen</code> (436 lines) and <code>app/territory.tsx</code> → <code>TerritoryScreen</code> (213 lines). Services: <code>heatRiskEngine</code> (389 lines), <code>heatProtocolService</code> (183 lines), <code>territoryEngine</code> (69 lines). Components: <code>HeatAlertBanner</code> , <code>HeatPulse</code> , <code>HeatRiskCard</code> , <code>MapLayerToggle</code> , <code>TerritoryMap</code> (stylized map per spec). OpenWeather proxied through API server with in-memory TTL cache + rate limiting. Tests: <code>territoryEngine.test.ts</code> . <code>Guardian</code> (<code>app/heat/guardian.tsx</code> → <code>GuardianHeatScreen</code> , 297 lines) is correctly feature-locked per spec: <code>guardian_intelligence_enabled</code> , <code>guardian_body_map_enabled</code> , <code>guardian_alerts_enabled</code> all false in production (<code>featureFlags/flags.ts</code>), true only in demo profile. Subscription gate ties Guardian Mode to Elite plan. No code changes required.
7	Share System + BECOME AForce footer	□	Audit: already fully shipped. Route <code>app/share.tsx</code> → <code>SharePreviewScreen</code> (577 lines). Screens: <code>MySharedStatusScreen</code> (162 lines). Components: <code>ShareCard</code> (170 lines), <code>ShareStory</code> (184 lines), <code>SharedStatusCard</code> (90 lines), <code>ShareJournalRecap</code> , <code>ShareText</code> . Services: <code>journalShareContext</code> , <code>journalShareCache</code> , <code>shareBroadcastEngine</code> , <code>shareTemplateEngine</code> , <code>shareAccentRotation</code> . BECOME AForce invite card lives in the Profile tab (referenced lines 483-496, rendered at 1342 + 1365), backed by <code>useGetMyReferralInfo</code> OpenAPI hook — server auto-issues code on first read, then returns code + totalClaims + tier + nextTier + claimsToNextTier (tier ladder). Referral spec #7 slices 1-3 (referral code generation, tier ladder, anonymous leaderboard) previously shipped. Tests: <code>journalShareCache.test.ts</code> , <code>journalShareContext.test.ts</code> , <code>shareAccentRotation.test.ts</code> , <code>shareBroadcastEngine.test.ts</code> , <code>shareTemplateEngine.test.ts</code> all green. No code changes required.
8	Legal + Compliance	□	Audit: already fully shipped. <code>app/legal/_layout.tsx</code> Stack hosts three routes — <code>terms.tsx</code> (Acceptance, What AForce OS is, Your account, Subscriptions, Acceptable use, Disclaimer of warranties, Limitation of liability, Changes), <code>privacy.tsx</code> (What we collect, How we use it, Where it lives, Your controls, Third parties Clerk/Stripe/ElevenLabs/OpenWeather, Children under 16, Changes), <code>health-disclaimer.tsx</code> (Performance not medicine, Talk to a professional first, Estimates have limits, In an emergency, You are in charge). All three render through shared <code>LegalDocumentScreen</code> (166 lines) with eyebrow/title/updatedAt/intro/sections/footer chrome. Both sign-in (line 176) and sign-up (line 291) link to all three legal pages. <code>legalSafetyService.ts</code> (94 lines) provides <code>impairmentFromBAC()</code> + <code>transportationPromptFor()</code> with mandatory "not a legal/medical determination" disclaimer for Social Mode safety; consumed by <code>socialModeEngine</code> and surfaced via <code>SocialSafetyCard</code> (severity-tinted accent). Tests: <code>bacEstimation.test.ts</code> (<code>describe('legalSafetyService')</code> block + "always returns disclaimer key" test). TestFlight checklist documents final-pass items (mailing address fill-in) for counsel review before public launch. No code changes required.
9	Feature Locks (Guardian / Clutch hidden)	□	Audit: flag infrastructure already shipped (<code>featureFlags/flags.ts</code> : <code>DEFAULT_FLAGS</code> has <code>clutch_access_enabled</code> , <code>clutch_heat_mode_enabled</code> , <code>clutch_inventory_enabled</code> , <code>clutch_clip_enabled</code> , <code>guardian_intelligence_enabled</code> , <code>guardian_body_map_enabled</code> , <code>guardian_alerts_enabled</code> all false ; <code>DEMO_ALL_ON_FLAGS</code> flips them ON for investor previews). Destination screens already wrap their bodies in <code><FeatureGate></code> (locked placeholder + "Activate Demo" CTA). Real gaps found and fixed: (1) <code>app/(tabs)/profile.tsx</code> <code>phaseEntryRow</code> rendered CLUTCH + GUARDIAN entry cards unconditionally (mounted twice at lines 1354 + 1375), exposing them in production navigation. Surgical fix: gated each card on its own flag (<code>showClutchEntry</code> / <code>showGuardianEntry</code>); row returns <code>null</code> when neither flag is on. (2) <code>screens/HeatRiskScreen.tsx</code> header "TEAM" button surfaced the Guardian roster route to consumer users. Surgical fix: added <code>useAppStore</code> import + <code>guardianEnabled</code> check; button only renders when <code>guardian_intelligence_enabled</code> is true. (3) Architect code review identified that <code>/heat/guardian</code> was still directly reachable via deep link even with both entries hidden. Surgical fix: added route-level <code><Redirect href="/heat" /></code> at the top of <code>GuardianHeatScreen</code> when <code>guardian_intelligence_enabled</code> is OFF (early return placed after all <code>useMemo</code> calls to respect Rules of Hooks). Triple defense in depth: entry hidden → surface gated → route bounced. Demo profile preserves full preview. Typecheck + 883/883 tests green.

#	Phase	Status	Notes
10	Investor Demo Overlay	☐	Built the scripted 60-second cinematic as SIX acts × 10s (Opening · Readiness 14→97 · HydroScan + voice · Social BAC shield · Territory/Heat Guard WARNING · The Standard sign-off), replacing the legacy 10-beat voice timeline. Everything is seeded from <code>data/demoProfile.ts</code> ; <code>services/demo/investorDemoBeats.ts</code> derives the cumulative beat schedule (<code>INVESTOR_DEMO_BEATS</code> , <code>beatAtMs</code> , <code>bandToLevel</code> , <code>new scoreToBand</code>). <code>components/investorDemo/InvestorDemoOverlay.tsx</code> renders a scene switch with tap-anywhere dismiss + auto-dismiss at exactly 60_000ms; only Act 3 speaks (via <code>commandSpeak</code>); NO store writes (Score-Protection — display-only projection of seeded numbers). New flag <code>demo_mode_enabled</code> (<code>false</code> in <code>DEFAULT_FLAGS</code> , <code>true</code> in <code>DEMO_ALL_ON_FLAGS</code>) + pure fail-closed <code>shouldShowInvestorDemo(flags, active)</code> gate both the <code>_layout.tsx</code> mount and the Profile launcher, so the overlay can never appear in production. Spec's "gold/lime" + "returns to welcome.tsx" superseded by the AForce Brand re-skin (brand red <code>#C1281B</code> + WHOOP state colors, no lime/gold) and the no-welcome-screen architecture (Modal close reveals the app underneath). Brand-color compliant; no nav/tab/dashboard or scoring-math changes. Tests: rewrote <code>investorDemoBeats.test.ts</code> for 6 acts + added <code>demoProfile.test.ts</code> (seed integrity) and <code>investorDemoGate.test.ts</code> (flag fail-closed) = 33 new/updated tests green; typecheck clean. TestFlight readiness captured in <code>docs/investor-demo-readiness.md</code> .

Addon Phases (AFORCE_SOCIAL_CRUISE_ADDON.md)

Locked until **all core phases above show** ☐ .

Social Additions

Phase	Status	Notes
Contexts	☐	Locked until core complete
Morning Reset	☐	Locked until core complete
Moments Engine	☐	Locked until core complete

Cruise Additions

Phase	Status	Notes
Voyage Recovery	☐	Locked until Social additions complete
Recovery Concierge	☐	Locked until Voyage Recovery complete
Cruise Contexts	☐	Locked until Recovery Concierge complete

Explicitly Not Built

- Recovery Journey — architecture only
- Journey Summary — architecture only
- Phantom — architecture only

☐ AFORCE_SOCIAL_CRUISE_ADDON.md

AFORCE OS — Social + Cruise Enhancement Layer (Addon)

Addon, not a rewrite. This document describes additions that layer on top of the core product defined in `AFORCE_FINAL_SPEC.md`. Do **not** merge any of this into the core spec. Do **not** start implementing any of this until all nine core phases have shipped and been approved.

Activation Order

After Phase 9 of the core spec is complete and approved:

1. Read this document.
2. Apply additions only — do not redesign Social Mode.
3. Apply additions only — do not redesign Cruise Mode.
4. Stop after completion.

Social Additions

Apply in order. Stop after each.

- **Contexts** — add Social Mode contexts (e.g. session types and modifiers) without altering the existing Social Mode surface.
- **Morning Reset** — additive morning ritual screen.
- **Moments Engine** — additive event/moment capture pipeline.

STOP after Social additions before starting Cruise.

Cruise Additions

Apply in order. Stop after each.

- **Voyage Recovery** — Cruise-mode recovery layer.
- **Recovery Concierge** — concierge surface that sits on top of Voyage Recovery.
- **Cruise Contexts** — Cruise-mode contexts, additive only.

STOP after Cruise additions.

Explicitly Out of Scope (never build under this addon)

- **Recovery Journey** — architecture only.
- **Journey Summary** — architecture only.
- **Phantom** — architecture only.

These three remain conceptual placeholders. Do not implement surfaces, services, routes, or storage for them.

Hard Rules

- Never modify existing Social Mode or Cruise Mode behavior — only add new contexts/screens/services alongside the existing ones.

- Never merge this document into `AFORCE_FINAL_SPEC.md` .
- Never start an addon until the core phase 9 is approved.
- One addition per session. Stop after each. Wait for approval.

📁 design/aforce-design-tokens.md

AForce OS Design Tokens — AForce Brand System

Source of truth: `artifacts/aforce-os/theme/*` **Figma import:** `design/aforce-tokens.json` (Tokens Studio for Figma, W3C format) **Version:** 2.1.0 — AForce Brand System

Design Philosophy

AForce OS uses the **AForce Brand System** — a cinematic dark performance aesthetic:

- **Near-black canvas** — solid backgrounds start at `#0D0D0D` , not pure black or dark gray
- **Content floats on darkness** — no visible card borders, structure comes from spacing
- **Signal Red hero** — AForce signal red `#C1281B` is the signature accent, used sparingly (thin lines, eyebrows, active states, CTAs)
- **Soursop green for positive status** — `#1FA35A` marks Peak / Optimal / success
- **Berry blue for secondary data** — `#1E5BFF`
- **Data-forward** — big numbers, small tracked labels, no decoration
- **Generous spacing** — when in doubt, add more whitespace
- **Soft glows, never hard shadows** — status colors radiate outward
- **Three type faces by role** — Archivo Black (display), IBM Plex Mono (eyebrows / metrics), Inter (body)

Color System Lock (v2.1.0)

The canonical AForce OS color system is the AForce Brand System. Near-black `#0D0D0D` canvas. Signal red `#C1281B` as the hero accent, used sparingly. Soursop green `#1FA35A` for positive status (Peak / Optimal / success). Berry blue `#1E5BFF` for secondary data and info. The score ladder runs Optimal green → Stable light-green → Declining amber → Risk orange → Critical red. Only completed behaviour changes score; accent color never implies score. Scrims (`rgba(0,0,0,a)`), drop-shadow color (`#000000`), and `text.inverse` (`#000000` , text on light/accent fills) stay pure black by design.

1. Colors

Backgrounds (near-black to near-invisible elevation)

Token	Hex	Usage
bg.primary	#0D0D0D	Screen background, canvas
bg.secondary	#050508	Slight elevation (barely visible)
bg.card	#0A0A0F	Card surfaces
bg.elevated	#101018	Elevated panels, sheets
bg.surface	#141420	Highest elevation (modals)
bg.overlay	rgba(0,0,0,0.92)	Fullscreen overlay scrim

Text

Token	Value	Usage
text.primary	#FFFFFF	Headlines, scores, primary content
text.secondary	rgba(255,255,255,0.55)	Body text, descriptions
text.muted	rgba(255,255,255,0.30)	Labels, metadata
text.ghost	rgba(255,255,255,0.18)	Placeholder, disabled
text.inverse	#000000	Text on light/accent backgrounds

Borders (near-invisible)

Token	Value	Usage
border.subtle	rgba(255,255,255,0.04)	Barely-there separators
border.medium	rgba(255,255,255,0.08)	Section dividers
border.strong	rgba(255,255,255,0.14)	Active/selected borders
border.accent	rgba(193,40,27,0.20)	Accent-tinted border

Fills (glass-on-black)

Token	Value	Usage
fill.light	rgba(255,255,255,0.02)	Barely-there card fill
fill.medium	rgba(255,255,255,0.05)	Default card fill
fill.strong	rgba(255,255,255,0.10)	Active/pressed fill

Hero Accent (Signal Red)

Token	Value	Usage
accent.primary	#C1281B	Primary accent, CTA, active states
accent.glow	rgba(193,40,27,0.50)	Button glow, accent halo
accent.dim	rgba(193,40,27,0.12)	Accent-tinted backgrounds
accent.subtle	rgba(193,40,27,0.06)	Very faint accent wash
accent.secondary	#1E5BFF	Berry blue, secondary data

Two band systems (intentional — do not merge)

The code runs **two parallel band systems** with different thresholds and roles:

- **Performance State (4 bands)** — `theme/colors.ts` `states` , classified by `utils/scoring/breakdown.ts` `resolveState` . Drives the **orb** (pulse / flare-on-peak / collapse-on-depletion), `riskTimer` , and

command selection. Thresholds: PEAK ≥ 90 , BALANCED ≥ 75 , RECOVERING ≥ 60 , else DEPLETED.

- **Score Status (5 bands)** — `theme/statusColor.ts` , mirrored by `utils/hydrationScore.ts` . Drives the **AI Coach status-color layer** (dots, borders, glows, CTA tint) and the score read-out. Thresholds: OPTIMAL ≥ 85 , STABLE ≥ 70 , DECLINING ≥ 50 , RISK ≥ 30 , else CRITICAL.

Both ladders share the same top green (`#1FA35A`) and bottom red (`#FF2800`); the middle bands differ by design.

Performance States (4 bands)

State	Primary	Glow	Dim
Peak (90-100)	<code>#1FA35A</code>	<code>rgba(31,163,90,0.50)</code>	<code>rgba(31,163,90,0.12)</code>
Balanced (75-89)	<code>#00E5C8</code>	<code>rgba(0,229,200,0.40)</code>	<code>rgba(0,229,200,0.12)</code>
Recovering (60-74)	<code>#FFA01E</code>	<code>rgba(255,160,30,0.40)</code>	<code>rgba(255,160,30,0.12)</code>
Depleted (0-59)	<code>#FF2800</code>	<code>rgba(255,40,0,0.40)</code>	<code>rgba(255,40,0,0.12)</code>

Score Status (5 bands)

Single source of truth: `theme/statusColor.ts` (mirrored by `utils/hydrationScore.ts`).

Band	Primary (calm)	Pressure Mode
Optimal (85-100)	<code>#1FA35A</code>	<code>#17C964</code>
Stable (70-84)	<code>#3DBE7A</code>	<code>#2BAA66</code>
Declining (50-69)	<code>#FFDE00</code>	<code>#FFC000</code>
Risk (30-49)	<code>#FF8C1A</code>	<code>#FF7A00</code>
Critical (0-29)	<code>#FF2800</code>	<code>#FF0040</code>

Wearable Integration Palette

The `whoop.*` token keys are retained for continuity with the wearable-snapshot surface; their values now follow the AForce Brand System (green status, blue strain). The provider brand swatch (`providers.whoop` = `#B6FF00`) keeps WHOOP's own lime so connect buttons render in the provider's real brand color.

Token	Value	Usage
<code>whoop.lime</code>	<code>#1FA35A</code>	Connected-status accent
<code>whoop.teal</code>	<code>#1E5BFF</code>	Strain bar fill
<code>whoop.recovery-green</code>	<code>#1FA35A</code>	Recovery $\geq 67\%$
<code>whoop.recovery-yellow</code>	<code>#FFDE00</code>	Recovery 34-66%
<code>whoop.recovery-red</code>	<code>#FF0026</code>	Recovery $\leq 33\%$
<code>whoop.panel-bottom</code>	<code>#0D0D0D</code>	Snapshot panel base
<code>whoop.ring-track</code>	<code>rgba(255,255,255,0.08)</code>	Ring background track
<code>whoop.strain-track</code>	<code>rgba(30,91,255,0.15)</code>	Strain bar background

Opacity Scale

Use for layering content on the near-black canvas:

`0.02` - `0.04` - `0.06` - `0.08` - `0.10` - `0.14` - `0.20` - `0.30` - `0.55` - `1.00`

2. Typography (Archivo Black · IBM Plex Mono · Inter)

Faces by role

Role	Family	Token	Usage
display	Archivo Black	fonts.display / roles.display	Hero numerals, wordmarks
eyebrow	IBM Plex Mono	roles.eyebrow	Tracked uppercase section labels
metric	IBM Plex Mono	roles.metric	Metric captions / values
mono	IBM Plex Mono	fonts.mono / roles.mono	Technical / tabular text
body	Inter (400 → 800)	fonts.regular ... fonts.bold	Everything else

Scale

Token	Size	Weight	Tracking	Usage
display-hero	80px	Bold	-1.5px	Hero score in cinematic view
display-score	64px	Bold	-1.5px	Score inside orb
display-mega	48px	Bold	-0.5px	Large feature numbers
display-title	36px	Bold	-0.5px	Screen titles
h1	28px	Bold	0	Section headings
h2	24px	SemiBold	0	Card headings
h3	20px	SemiBold	0	Subheadings
body-lg	17px	Medium	0	Primary body
body-base	15px	Regular	0	Default body
body-sm	13px	Regular	0	Secondary body
caption	11px	Medium	0	Metadata
eyebrow	11px	Bold	3px	Section labels (UPPERCASE)
eyebrow-sm	9px	Bold	3px	Tiny labels (UPPERCASE)
metric-label	9px	SemiBold	2px	Metric labels (UPPERCASE)
metric-value	24px	Bold	-0.5px	Metric numbers
pill	10px	Bold	1px	Pill/tag text (UPPERCASE)

3. Spacing

0 - 4 - 8 - 12 - 16 - 20 - 24 - 28 - 32 - 40 - 48 - 56 - 64 - 80 - 96

Spacing rule: sections should have 40-64px between them. Cards should have 20px internal padding. Never let elements feel crowded.

4. Radii

Token	Value	Usage
<code>none</code>	0	Sharp edges (rare)
<code>sm</code>	8px	Small chips, pills
<code>md</code>	12px	Cards, inputs
<code>lg</code>	16px	Metric cards, sections
<code>xl</code>	20px	Large cards
<code>2xl</code>	24px	Sheets, modals
<code>3xl</code>	32px	Hero containers
<code>full</code>	9999px	Circles, rounded pills

5. Component Dimensions

Component	Token	Value
Orb	<code>orb-size</code>	200px
Orb ring stroke	<code>orb-stroke</code>	6px
Orb glow blur	<code>orb-glow-radius</code>	32px
Recovery ring	<code>ring-size</code>	132px
Recovery ring stroke	<code>ring-stroke</code>	8px
Strain bar height	<code>strain-bar-height</code>	6px
CTA button height	<code>cta-height</code>	56px
CTA radius	<code>cta-radius</code>	14px
Status pill height	<code>pill-height</code>	28px
Share button	<code>share-btn-size</code>	36px
Content padding	<code>content-padding</code>	20px

6. iPhone 14 Pro Layout

Constant	Value
Screen size	393 x 852
Status bar	54px
Safe area top	59px
Safe area bottom	34px
Tab bar	84px
Content padding	20px each side
Usable content width	353px

7. Shadows / Glows

AForce never uses hard box shadows. Everything is a soft radial glow that matches the status color:

Token	Color	Blur	Usage
orb-glow	rgba(31,163,90,0.50)	40px	Orb ambient glow (peak)
glow-peak	rgba(31,163,90,0.35)	24px	Peak state elements
glow-balanced	rgba(0,229,200,0.25)	18px	Balanced state
glow-recovering	rgba(255,160,30,0.30)	14px	Recovering state
glow-depleted	rgba(255,45,85,0.40)	12px	Depleted state
cta-glow	rgba(193,40,27,0.20)	16px	CTA button ambient

How to Use in Figma

1. Open Tokens Studio plugin (Cmd+P, type "Tokens Studio")
2. Import `aforce-tokens.json` (three-dot menu, Tools, Load from file)
3. Push to Figma Variables (Tools, Export to Figma Variables)
4. Every color, font size, spacing value becomes a Figma Variable
5. When building frames, use variables for all fills/strokes/text
6. When the codebase changes, ask Replit to re-export, then re-import -- everything stays in sync

artifacts/aforce-os/docs/social-mode-safety-spec.md

Social Mode — Safety & Estimation Spec

This document is the canonical reference for the BAC estimator, impairment escalation, transportation prompts, and recovery flow that power **AForce OS Social Mode**. Anyone touching the engine, UI, or copy must read this first.

1. Voice & tone

Social Mode is **calm, protective, never preachy**.

- The system **never** moralizes, lectures, or shames the user for drinking. It does not use words like "stop drinking", "you should not", or "irresponsible".
- The system **never** says the user is "safe to drive" or implies legal compliance. The strongest positive language allowed is "your estimate is currently low" — it never crosses into permission.
- When escalating, the system uses protective verbs: *plan a ride, switch to water, arrange safe transport, stop alcohol intake*. It describes outcomes, not character.
- Headlines are short. Bodies are 1–2 sentences max.

2. BAC estimation (Widmark approximation)

Implemented in `services/bacEstimationService.ts`.

```
gramsAlcohol = sum( drink.oz * (drink.abv/100) * 0.789 * 29.5735 )
bodyMassG    = bodyWeightLbs * 453.592
r             = 0.68 (male / unspecified) | 0.55 (female)
foodFactor    = 0.92 if ateRecently else 1
bacRaw        = (gramsAlcohol / (bodyMassG * r)) * 100 * foodFactor
elapsedH      = hours since first drink
bacCurrent    = max(0, bacRaw - 0.015 * elapsedH)
```

The output is a **range** widened by ± 0.01 around the point estimate. We never display a single false-precision number.

Field	How it's derived
<code>rangeLow</code> / <code>rangeHigh</code>	Point estimate ± 0.01 .
<code>trend</code>	Compare current BAC to BAC 15 min ago; $ \Delta < 0.005$ = steady .
<code>confidence</code>	high if sex is provided AND $\geq 50\%$ of drinks have explicit oz/abv AND ≤ 8 drinks.
<code>timeToClearMinutes</code>	$(\text{bacCurrent} - 0.005) / 0.015 * 60$, rounded up to nearest 5 min.

Inputs

- `drinks` : { type, loggedAt, abv?, oz? }[] — abv / oz fall back to the catalog defaults when omitted.
- `bodyWeightLbs` : floored to 80 lbs to avoid divide-by-tiny errors.
- `sex` : optional. 'male' | 'female' | 'unspecified' .
- `ateRecently` : optional boolean.

Drink catalog (`data/alcoholDrinks.ts`)

Type	Default oz	Default ABV	Decay multiplier	Sugar load
beer	12	5.0	1.15	3
wine	5	12.5	1.20	4
cocktail	8	14.0	1.30	8
liquor	1.5	40.0	1.35	1
hard_seltzer	12	5.0	1.15	1
custom	6	12.0	1.25	4

3. Impairment escalation matrix

Implemented in `services/legalSafetyService.ts` . Mapping uses the **midpoint** of the BAC range so trend is smooth across refreshes.

BAC midpoint	Level	Safety card	Stop drinking?	Coach command
< 0.030	LOW	hidden	no	(standard hydration / pacing copy)
0.030 – 0.049	ELEVATED	hidden	no	(standard hydration / pacing copy)
0.050 – 0.079	MODERATE	shown — caution	no	"Plan a ride before your next drink"
0.080 – 0.119	HIGH	shown — warning	yes	"Do not drive. Use a rideshare."
≥ 0.120	CRITICAL	shown — critical	yes	"Stop alcohol intake. Recovery req."

The safety card is hidden at `LOW` and `ELEVATED` so the user only sees the legal-protection language when it actually applies. The "stop drinking" sub-prompt only appears at `HIGH/CRITICAL`.

4. Disclaimer policy

Every surface that shows a BAC value, impairment level, transportation prompt, or recovery time **must** render the standard disclaimer pair:

Estimate only · Not a legal or medical determination.

The i18n keys are `social.estimate_only` and `social.not_legal_medical`. They are translated in all 6 supported locales (`en`, `es`, `fr`, `de`, `pt`, `it`).

The system **never**:

- Tells the user they are "safe to drive".
- Promises a specific BAC at a specific future time.
- Asserts compliance with any legal limit (those vary by jurisdiction and by individual physiology).
- Provides medical advice (kidney/liver concerns, medication interactions, etc.).

5. Recovery Mode

Triggered when the user taps **End Night** in the Social Mode sheet. Engine sets `socialMode.endedAt` and the rollout flips `inRecoveryWindow = true` for the next 8 hours (`RECOVERY_WINDOW_MS = 8 * 60 * 60 * 1000` in `socialModeEngine.ts`).

While in recovery the UI renders `RecoveryModeCard` showing:

1. The estimated time until BAC clears (from `bac.timeToClearMinutes`).
2. A 3-step morning protocol — water → AForce RTD → 7+ hours sleep.
3. The standard disclaimer pair.

The home banner shifts to amber and reads `social.recovery_active` .

6. Orb overlay

`StatusPulseOrb` accepts `socialOverlay?: { alcoholLoad: number; unstable: boolean }` .

- `alcoholLoad` $\in [0, 1]$ is derived from the active decay multiplier and pushes a subtle violet outer ring.
- `unstable = true` when impairment is `HIGH` or `CRITICAL` — the ring flips crimson and pulses faster.

The overlay is purely additive; it never replaces the normal hydration gradient.

7. Test invariants

services/_tests_/bacEstimation.test.ts pins:

- Zero drinks → zero BAC, zero clear time.
- One beer → LOW band.
- Four quick liquor shots → at least MODERATE.
- Trend: rising right after drinks, falling after long elimination.
- Time-to-clear is non-negative and a multiple of 5 minutes.
- Confidence degrades when sex is unspecified.
- Food intake softens the BAC estimate vs an empty stomach.
- Safety prompt is hidden at LOW/ELEVATED.
- Safety prompt escalates to "do not drive" at HIGH and CRITICAL.
- Safety disclaimer key is always returned.

These invariants must continue to hold after any change to the engine or the impairment thresholds.

artifacts/aforce-os/docs/investor-demo-readiness.md

AForce OS — Investor Demo Readiness

Last updated: June 22, 2026 · Phase 10 (Investor Demo Overlay) complete.

This is the single-page status for the **60-second investor demo** and the path from "runs in the room today" to "in testers' hands" to launch. It is a companion to `docs/TESTFLIGHT_CHECKLIST.md` (the build/submit runbook) — read that for the step-by-step Apple mechanics.

Ready now (works today, in this build)

- **The 60-second demo plays end-to-end.** Six acts × 10 seconds:
 1. **Opening** — AForce wordmark + "The Performance Operating System."
 2. **Readiness Score** — orb climbs Depleted → Peak, score animates 14 → 97.
 3. **HydroScan** — product recognition (AForce Hydration Stick · OPTIMAL MATCH)
 - the AI Voice Engine speaks "You're back in range. Lock in."
 4. **Social Mode** — BAC safety overlay, crimson ring on the orb.
 5. **Territory + Heat Guard** — stylized map sector + Heat Guard → WARNING.
 6. **The Standard** — clean Peak orb + "Built for people who don't get to be off."
- **Tap anywhere to skip**, or it **auto-dismisses at exactly 60s** and returns to the app underneath.
- **Launch path:** Profile → Voice Coach section → ► **LAUNCH INVESTOR DEMO · 60s** (only visible when the demo flag is on — see below).
- **Score-Protection holds:** the overlay is a display-only projection of the seeded numbers in `data/demoProfile.ts`. It never awards, mutates, or persists score, never reads live user state, and writes nothing to the store.
- **Brand-correct:** AForce brand red (`#C1281B`) + WHOOP recovery state colors. No lime / gold (the spec's original gold/lime wording predates the brand re-skin and is intentionally superseded).

Done this sprint (Phase 10)

- Replaced the legacy 10-beat voice script with the spec's **6-act, 10s-each** cinematic, driven entirely by a single seed (`data/demoProfile.ts`).
 - `services/demo/investorDemoBeats.ts` derives the playable schedule (cumulative start times, `beatAtMs` , `bandToLevel` , `new scoreToBand` for the continuous orb re-tint during the climb).
 - Rewrote `components/investorDemo/InvestorDemoOverlay.tsx` into a per-act scene switch: opening / readiness / hydroScan / social / territoryHeat / standard, with tap-anywhere + 60s auto-dismiss, reanimated orb/voice-halo/scan-line/ social-ring/progress drivers, and voice on Act 3 only via `commandSpeak` .
 - Added the `demo_mode_enabled` feature flag (OFF in `DEFAULT_FLAGS` , ON in `DEMO_ALL_ON_FLAGS`) and the pure, fail-closed `shouldShowInvestorDemo(flags, active)` helper.
 - **Gated both entry points** on the flag: the `_layout.tsx` overlay mount and the Profile launcher. Production builds (flag OFF) have no path to the overlay.
 - Tests: rewrote `investorDemoBeats.test.ts` for 6 acts and added `demoProfile.test.ts` (seed integrity) + `investorDemoGate.test.ts` (production-safe / fail-closed gating) — **33 tests, all green**; typecheck clean.
 - Small TestFlight fix: declared `ITSAAppUsesNonExemptEncryption: false` in `app.json` (standard HTTPS only) to avoid the "Build is missing compliance" stall on upload.
-

Before TestFlight (internal beta) — owner / Apple actions

These are **account / credential** tasks, not code. Code side is ready.

- ☐ **Apple Developer Program** enrollment active.
- ☐ **Fill the two iOS submit IDs.** `eas.json` still has placeholders (`REPLACE_WITH_APP_STORE_CONNECT_APP_ID` , `REPLACE_WITH_APPLE_TEAM_ID`). Use the helper (never hand-edit): from repo root, `EAS_ASC_APP_ID=... EAS_APPLE_TEAM_ID=... pnpm --filter @workspace/scripts run eas-configure-submit` .
- ☐ **Privacy policy live at a real URL.** `legal/privacy-policy.md` exists but needs the mailing address filled in, counsel review (HealthKit + CCPA + GDPR), and publishing at a stable URL for App Store Connect.
- ☐ **Backend reachable from a non-Replit network** (api-server deployed) so a real device build can sign in and load state.
- ☐ **EXPO_PUBLIC_* production env vars** baked into the EAS build (Clerk publishable key, API base URL). Missing ones are the #1 cause of TestFlight-only crash-on-launch.
- ☐ Build + submit: `pnpm --filter @workspace/aforce-os run eas:build:ios then ... eas:submit:ios` (see `docs/TESTFLIGHT_CHECKLIST.md`).

Already green for TestFlight (verified in-repo):

- `app.json` `version` `1.0.0` , `ios.bundleIdentifier` `com.aforce.os` , `buildNumber` set.
 - App icon is **1024×1024 PNG, RGB (no alpha)** — Apple's hard requirement.
 - All `infoPlist` permission strings are human-written (camera, location, HealthKit r/w).
 - `eas.json` has complete development / preview / production profiles.
 - Typecheck passes.
-

Before Miami (October field demo)

- ☐ Run the demo on a **physical iPhone** (not the simulator / web preview) — HealthKit, haptics, and the Voice Engine's audio path only fully exercise on device.

- ☐ Confirm the **ElevenLabs voice** plays on Act 3 on the venue network (Act 3 is the only act that speaks; everything else is silent by design).
 - ☐ Pre-flight the venue: airplane-mode fallback. The demo itself is self-contained (seeded, no network needed to render), but the Act-3 voice uses the ElevenLabs proxy — confirm behavior if Wi-Fi is flaky.
 - ☐ Decide the **launch surface** for the room: keep it behind the Profile launcher, or wire a faster trigger for the presenter (flag stays the gate).
 - ☐ Capture a screen-recording of a clean run as a backup if live demo fails.
-

Before public launch (January)

- ☐ `demo_mode_enabled` **MUST stay false** in the production flag set (`DEFAULT_FLAGS`). This is the hard rule — verified by `investorDemoGate.test.ts` . The overlay must never ship visible.
 - ☐ Full App Privacy questionnaire + age rating in App Store Connect.
 - ☐ App Store screenshots from a real device/simulator binary (Replit web preview cannot produce submission-grade assets — see `replit.md`).
 - ☐ Counsel sign-off on privacy policy + health disclaimer for production.
 - ☐ External TestFlight (Beta App Review) pass before broad distribution.
-

Risk level

Low (for the demo itself). The overlay is self-contained, seeded, gated, and Score-Protection-safe; it cannot leak into production or corrupt user state. The only runtime dependency is the Act-3 ElevenLabs voice (graceful if it fails to play — the visuals carry the act).

The remaining risk is operational, not code: Apple account setup, the two submit IDs, a published privacy policy, production env vars in the EAS build, and a physical-device dry run before any live audience. None block the demo running in the room today; all block getting it onto a stranger's phone.

Next sprint (suggested, not committed)

- Wire a presenter-friendly trigger (e.g. a long-press affordance) behind the same flag for faster on-stage launches.
 - Add a reduced-motion variant of the overlay (the rest of the app is reduced-motion aware; the demo currently always animates).
 - Optional: a short "loop" mode for an unattended booth (auto-replay), still flag-gated and still Score-Protection-safe.
-

**📁 artifacts/aforce-
os/docs/TESTFLIGHT_CHECKLIST.md**

AForce OS — TestFlight Submit Checklist

A step-by-step runbook for getting an internal beta build of AForce OS into TestFlight for the team. Estimated total time on a fresh setup: **3-6 hours** (most of it waiting on Apple).

Phase 0 — Prerequisites (do once)

0.1 Apple Developer Program

- ☐ Enroll at <https://developer.apple.com/programs/> (99/yr individual, 299/yr organization).
- ☐ Wait for approval (usually same-day for individuals, up to a few business days for organizations).
- ☐ Confirm enrollment is active in [App Store Connect](#).

0.2 Expo / EAS account

- ☐ Sign up at <https://expo.dev/signup> if you don't have one.
- ☐ On your local machine: `npm install -g eas-cli`
- ☐ `eas login` and confirm with `eas whoami`.

0.3 Privacy policy live at a real URL

- ☐ Take `legal/privacy-policy.md`, fill in the mailing address and any TBD fields.
 - ☐ Have it reviewed by counsel (HealthKit + CCPA + GDPR).
 - ☐ Publish at a stable URL, e.g. <https://drinkaforce.com/privacy>. (A Notion public page or a static HTML file works for beta; you'll need a real domain page for App Store production.)
 - ☐ Save the URL — you'll paste it into App Store Connect.
-

Phase 1 — App Store Connect setup (do once)

1.1 Create the app record

- ☐ Go to App Store Connect → **My Apps** → + → **New App**.
- ☐ Platform: **iOS**
- ☐ Name: **AForce OS** (must be unique on the App Store)
- ☐ Primary Language: English (U.S.)
- ☐ Bundle ID: `com.aforce.os` — if it doesn't appear in the dropdown, register it first at developer.apple.com → **Certificates, Identifiers & Profiles** → **Identifiers** → +.
- ☐ SKU: `AFORCE-OS-001` (any unique string)
- ☐ User Access: **Full Access**
- ☐ Click **Create**.

1.2 Fill in App Information

- ☐ **Privacy Policy URL** → paste the URL from 0.3
- ☐ **Category** → Primary: Health & Fitness; Secondary: Lifestyle (or your choice)
- ☐ **Content Rights** → declare whether the app contains third-party content
- ☐ **Age Rating** → walk through the questionnaire (likely 4+)

1.3 App Privacy questionnaire (App Store Connect → App Privacy)

This is required before TestFlight external testing and before App Store submission. For internal-

only TestFlight you can defer it, but it's faster to do it now.

Declare data collection for:

- ☐ **Health & Fitness** — used for App Functionality, **not** linked to user identity if you keep it client-side; linked if you sync to your backend
- ☐ **Contact Info → Email** — used for App Functionality (auth)
- ☐ **Identifiers → User ID** — used for App Functionality
- ☐ **Usage Data → Product Interaction** — used for Analytics
- ☐ **Diagnostics → Crash Data, Performance Data** — used for App Functionality

For each category, confirm: data is **not used for tracking** and **not shared with third parties for advertising**.

Phase 2 — Pre-build sanity checks

Run these before every build.

- ☐ `pnpm --filter @workspace/aforce-os run typecheck` passes
 - ☐ `app.json` `version` is correct (currently `1.0.0`)
 - ☐ `app.json` `ios.bundleIdentifier` is `com.aforce.os` and matches App Store Connect
 - ☐ All `infoPlist` permission strings read like a human wrote them (Apple rejects placeholder text)
 - ☐ App icon at `assets/images/icon.png` is **1024 × 1024 PNG, no transparency, no alpha channel** (Apple's hard requirement)
 - ☐ Smoke-test on a real iPhone: sign in, log a drink, scan a barcode, grant HealthKit, view recovery score, sign out. Simulator does **not** support HealthKit, so a physical device is required.
 - ☐ No `console.log` of secrets, no hard-coded test API keys, `EXPO_PUBLIC_*` env vars set for production
 - ☐ Backend (api-server) is deployed and reachable from a non-Replit network
-

Phase 3 — Build with EAS

From the repo root.

3.1 First-time only: configure credentials

- ☐ `cd artifacts/aforce-os`
- ☐ `eas build:configure` — confirms `eas.json` (already present) and links to your Expo project
- ☐ `eas credentials` → iOS → **Set up a new distribution certificate**. EAS can manage certificates and provisioning profiles for you; say yes.

3.2 Build for TestFlight

- ☐ `eas build --profile production --platform ios`
- ☐ Wait ~15-30 minutes for the build to finish on EAS's macOS workers
- ☐ Build artifact (`.ipa`) appears in your Expo dashboard

The `production` profile in `eas.json` is the right one for TestFlight — `distribution` defaults to "store" (App Store Connect), which is what TestFlight reads from. The `preview` profile (`distribution: internal`) is only for ad-hoc IPAs you sideload outside TestFlight.

Phase 4 — Submit to TestFlight

4.1 Upload

- ☐ `eas submit --profile production --platform ios`
- ☐ Pick the build from the previous step
- ☐ Provide your Apple ID and an [app-specific password](#) (or use API key — recommended for CI)
- ☐ Wait for upload + Apple's processing (~15–60 minutes; you'll get an email)

4.2 Configure the TestFlight build (App Store Connect → TestFlight tab)

- ☐ Once Apple finishes processing, the build appears under **iOS Builds**
- ☐ Click the build → fill in **Test Information**:
 - **What to Test**: "First internal build of AForce OS. Please test: sign-in, hydration logging, barcode scan, HealthKit grant flow, recovery score, sign-out. Report bugs to bburrell@alkalineforce.com."
 - **Beta App Description**: short paragraph from the pitch deck
 - **Email**: bburrell@alkalineforce.com
 - **Privacy Policy URL**: same as in Phase 1.2
- ☐ **Export Compliance**: answer the encryption question. AForce OS uses standard HTTPS only → choose **"Uses standard encryption exempt from export compliance"** (ITSAppUsesNonExemptEncryption = false).

4.3 Add internal testers (no Apple review needed)

- ☐ App Store Connect → TestFlight → **Internal Testing** → + → create a group called "AForce Team"
- ☐ Add team members by email (must have an App Store Connect role on your team — invite them via **Users and Access** first)
- ☐ Enable the build for the group
- ☐ Up to **100 internal testers**, each on up to 30 devices, no Apple review

4.4 (Optional) Add external testers (light Apple review)

- ☐ TestFlight → **External Testing** → + → create a group like "Friends & Family Beta"
- ☐ Add testers by email — they don't need an App Store Connect account
- ☐ Submit the build for **Beta App Review** (usually 1–2 business days)
- ☐ Apple will check: app launches, HealthKit usage matches the declared purpose, no broken core flows
- ☐ Up to **10,000 external testers**, each build is valid for 90 days

Phase 5 — Iterate

For each new beta build:

- ☐ Bump `ios.buildNumber` in `app.json` (or rely on `appVersionSource: "remote"` in `eas.json`, which auto-increments — already set)
 - ☐ `eas build --profile production --platform ios`
 - ☐ `eas submit --profile production --platform ios`
 - ☐ Update **What to Test** with the changes since the last build
 - ☐ Internal testers get the build immediately; external testers get it after Apple's quick re-check (subsequent reviews are usually <24h)
-

Common gotchas

Symptom	Fix
Build fails: "Missing privacy manifest"	Add <code>app.json</code> → <code>ios.privacyManifests</code> or rely on Expo SDK 54+ defaults; check Expo release notes for your SDK
App Store Connect: "Invalid Bundle. The bundle identifier cannot be registered"	Someone else already registered <code>com.aforce.os</code> . Pick a new one and update <code>app.json</code>
TestFlight: "Build is missing compliance"	Export Compliance not answered (Phase 4.2)
HealthKit prompt never appears	You're testing in the simulator. Use a real device
Crash on launch in TestFlight only	Almost always a missing <code>EXPO_PUBLIC_*</code> env var that's set locally but not baked into the production build. Set it in EAS: <code>eas secret:create</code>
Apple rejects: "Health data used for advertising"	Your privacy policy or App Privacy answers are inconsistent. HealthKit data must never be marketed-as-used for ads

Useful links

- [EAS Build docs](#)
- [EAS Submit docs](#)
- [TestFlight overview](#)
- [App Store Connect](#)
- [Apple's HealthKit review guidelines](#)

📄 docs/competitive-moat.md

AForce — Competitive Moat & Defensibility

⚠ **Internal strategy & positioning document — not user-facing or marketing copy.** All "prescribe" / "prescription" language in this document is **internal framing only**. Any user-facing, in-app, or marketing use must follow the v1 language lock in `validation-methodology.md` §6 (Founder Decision, 2026-06-01): only **"recommendation"** is permitted — never "prescription." HydroScan and every surface stay advisory.

CPG is copied. Wearables are commoditized. Apps are forgotten. AForce is a **closed-loop performance OS** that ships a consumable. The category has drinks. AForce has a loop.

This document maps every AForce surface against every relevant incumbent and shows the structural reason each competitor cannot take that surface from us. It is the long-form companion to deck slides 17 (Competitive Advantage) and 18 (Competitors) — extended beyond the beverage category into wearables, smart bottles, pro-team SaaS, aggregators, and hospitality.

1 · The Structural Thesis

Every entrenched player owns **one layer**. AForce is the only company stacking all four:

Layer	Who owns it today	What they can't do
Sensor (HR, HRV, sleep, strain)	WHOOP · Oura · Apple Watch · Garmin	Ship a consumable. Close the loop with intake.
Consumable (electrolytes, hydration)	Gatorade · Liquid IV · LMNT · Nuun	See the body. Personalize. Coach. Retain via software.
Vessel (smart bottle)	Hidrate Spark · HidrateSpark Pro	Influence physiology — they only count ounces in a cup.
Aggregator (Apple Health · Google Health Connect · Samsung Health · MyFitnessPal)	Apple · Google · Samsung · Under Armour	Prescribe. They are read-only mirrors with no opinion and no SKU.

AForce occupies the **only** position where the sensor signal, the prescription, the consumable that fulfills the prescription, and the verification of compliance are all in one product. That is the loop. Every additional day of usage compounds the loop's accuracy and the user's switching cost.

Single-line moat: *No CPG brand has built the OS. No SaaS brand has the can. No wearable has the SKU. No bottle has an opinion.*

2 · Competitor-by-Competitor Defense

2.1 WHOOP

Their strength: Best-in-class strain & recovery coaching. Subscription-native. Coach-trusted in pro sports. **Where they hit AForce:** Recovery overlay on Home; Connected Devices integration; pro-team Guardian/Clutch positioning. **Why they can't take it:**

- WHOOP tells you *you're depleted*. AForce tells you *drink this stick now, recheck in 12 minutes, then this RTD at the half*. Prescription beats observation.
- WHOOP has no fulfillment. Every recommendation is a dead-end — "rest more, hydrate more" with no SKU, no cart, no compliance loop.
- The **AForce OS Recovery overlay** ingests WHOOP as one of seven sources (`biometricsAggregator.ts`). We turn their data into *our* prescription. They cannot reverse the integration without becoming a CPG company, which their hardware-subscription P&L cannot absorb.
- Pro-team penetration: WHOOP sells **straps**. AForce sells **Clutch (live command grid) + Guardian (composite risk score) + the consumable the staff already buys**. A coach can't pour a strap into a player.

AForce defense: Position WHOOP as a *premium upstream sensor* in our integration tier — they make our prescription sharper. We make their data actionable.

2.2 Oura

Their strength: Sleep & HRV gold standard. Strong female-health and longevity narrative.

Premium hardware aesthetic. **Where they hit AForce:** Recovery overlay; sleep-quality inputs to compliance scoring; premium dark UX. **Why they can't take it:**

- Oura's daily Readiness score is a **diagnosis without a treatment**. Users wake up, see "62 — pay attention," and have nowhere to act. AForce reads the same HRV and routes it directly into a 16-oz + 1 stick prescription with a 15-minute recheck.
- Oura has no daytime loop. The ring reports nightly — AForce reports **every sip, every heat-index spike, every quarter break**.
- We share Oura's design language (premium, dark, restrained) but layer **operational density** on top: AForce cinematic visualization, band-colored scores, live cadence. Oura looks like a journal. AForce looks like mission control.

AForce defense: Oura users self-select for *willingness to optimize*. They are our highest-LTV cohort. Connect via Oura's HRV/Readiness API → we own the next 16 hours of their day.

2.3 Apple (Apple Health + Apple Watch + Apple Fitness+)

Their strength: Default ecosystem. Distribution. Trust. Free baseline metrics. iCloud-grade privacy story. **Where they hit AForce:** They sit underneath everything. They could surface a "hydration" vertical at any WWDC. **Why they can't take it:**

- Apple is a **horizontal platform**. They will never sell a stick, prescribe a sodium-band, or build a Heat Guard rule that pulls a player from rotation. Vertical specialists win the deep customer.
- Apple Health is **an aggregator with no opinion**. AForce is the opinion *on top* of Apple Health. We are a complement, not a substitute — we read from HealthKit and write back.
- Apple cannot do **Cruise Mode, Clutch, or Guardian** — those are vertical B2B/B2B2C surfaces with bespoke domain logic (heat-index, alcohol+sun risk, NBA/college roster monitoring). Apple's playbook is to expose primitives, not to operate vertical SaaS.
- Watchface real estate is the only meaningful Apple risk. Mitigation: ship a **first-class watchOS complication** that surfaces the live AForce score → we colonize their glass instead of being colonized.

AForce defense: Be the *best HealthKit citizen in the hydration vertical*. Apple cannot punish a vertical that respects the platform.

2.4 Garmin (Garmin Connect)

Their strength: Endurance-athlete loyalty. GPS workouts. Best-in-class battery. Hydration logging already in the watch. **Where they hit AForce:** Sweat-rate calculator territory; endurance-athlete persona overlap. **Why they can't take it:**

- Garmin's hydration log is **a counter, not a coach**. It asks the user to enter ounces. AForce computes the prescription from preKg/postKg/duration/fluidIn/urineOut using **ACSM Sawka 2007** and routes it into a four-band sodium target from **Baker 2017** (see [ScienceScreen.tsx](#)). That's the difference between a tally sheet and a sports-medicine engine.
- Garmin Connect's UI is a 2014-era dashboard. AForce ships a 2026 dark cinematic OS calibrated to elite-aesthetic expectations.
- Garmin has no consumable. Our **Sweat Calculator → Store** funnel converts a free measurement into a recurring SKU. Garmin would have to become a CPG company.

AForce defense: Position Sweat Calculator as the *open methodology* (Quick / Precision / Estimate modes) Garmin athletes use to validate Garmin's auto-tracking — and discover AForce's prescription.

2.5 Gatorade / Gatorade Gx (incl. Gx Sweat Patch)

Their strength: \$6B brand. Universal availability. PepsiCo distribution. The Gx Patch is the only mainstream sweat-sodium sensor. **Where they hit AForce:** Hydration-with-electrolytes category; sweat-sodium personalization claim; pro-team locker rooms. **Why they can't take it:**

- The Gx Patch is a **one-shot diagnostic** — single-use, \$25, sodium concentration only, no daily loop, no software memory. AForce ingests that result (or its own preKg/postKg input) and reuses it across every workout, every heat day, every cruise day, forever.
- Gatorade is a **beverage company that ships an app**. AForce is a **software company that ships a beverage**. Org-chart determines roadmap: Gatorade ships flavors; AForce ships engine improvements.
- Gx software has no Recovery overlay, no Heat Guard, no Guardian risk score, no Clutch command grid, no Cruise Mode. Eight surfaces vs one.
- **Premium positioning:** Gatorade is mass-market. AForce is **alkaline pH 8.8+, functional superfoods, four formats, \$34.99 / 12-ct sticks** — a price point Gatorade structurally cannot reach without cannibalizing the parent brand.

AForce defense: Let Gatorade own the sideline cooler. We own the **prescription that decides what goes in the cooler**.

2.6 Liquid IV / LMNT / Nuun (Electrolyte Brands)

Their strength: Direct-to-consumer mastery. Strong founder/community brands (LMNT especially). Clean-ish ingredient stories. **Where they hit AForce:** The Store screen; subscription pricing; "hydration multiplier" claim. **Why they can't take it:**

- All three are **single-format SKUs in a paper sleeve**. AForce ships **sticks + cans + bundles + subscriptions** under one brand, with the **OS routing the user to the right format for the right context** ("Berry Blast now, maintain PEAK" vs "Watermelon Surge for HEAT recovery"). They sell powder. We sell a **prescribed system**.
- None of them have a wearable integration, a sweat calculator, an achievement system, or a Heat Guard rule. The Store is the *output* of the OS — they only have the output, with no engine behind it.
- LMNT is closest in tone (premium, founder-led) but is a **single-flavor sodium-forward SKU**. AForce's **alkaline + functional + AI-coached** stack is structurally broader.
- **Subscription stickiness:** Liquid IV's churn is high because the value prop ends at the cup. AForce subscription compounds with **streaks, achievements, Recovery windows, Cruise Mode, Clutch access** — every additional surface raises switching cost.

AForce defense: Win the customer once on a stick → onboard them into the OS → 6 months later they cannot leave without losing their Streak, their Achievements, and their personalized Recovery curve.

2.7 Hidrate Spark (Smart Bottle)

Their strength: First-to-market smart-bottle category leader. Glow ring. Bluetooth sync. **Where they hit AForce:** Compliance verification ("did you actually drink it"); intake logging. **Why they can't take it:**

- Hidrate Spark **counts ounces**. It cannot tell you *what* to drink, *why*, or *when in your strain cycle*. It is a peripheral with no engine.
- Hidrate has no consumable. Their bottle is empty when you buy it — the customer goes to **us** to fill it.
- AForce can integrate Hidrate as a **PHANTOM-class verifier** (Profile screen has a

PHANTOM Band toggle and a CLUTCH Clip — same architecture extends to a smart-bottle peripheral). They become a sensor in our loop, not a competitor to it.

- Their app retention is single-digit because counting water is a six-month novelty. Our retention compounds because **we're a coach, not a counter**.

AForce defense: Ship a **Bring-Your-Own-Bottle** integration tier. Hidrate becomes a Profile-screen device card alongside Apple Watch Ultra and Oura Ring.

2.8 Catapult / Kitman Labs / VALD (Pro-Team Sports Science SaaS)

Their strength: Deep enterprise penetration in NBA, NFL, EPL, NCAA. GPS load monitoring.

Force-plate biomechanics. Trusted by performance staff. **Where they hit AForce:** Clutch

("Command the Team") and Guardian ("Protect the Roster") screens. **Why they can't take it:**

- Catapult tracks **external load** (sprints, accelerations). VALD tracks **neuromuscular readiness**. Kitman aggregates training data. **None of them touch hydration, sodium, heat-index, or the consumable that resolves the deficit**. AForce is the missing layer — and the only layer with a SKU attached to every recommendation.
- Their UIs are **pre-2020 enterprise dashboards** built for a sports scientist with a laptop in a meeting room. AForce ships a **mobile-first, glance-readable, color-coded command grid** built for an assistant coach with 11 seconds during a timeout. (Screenshot: live grid with PF/SG/PG color dots, hydration scores, "12 ounces + 1 stick at next dead ball.")
- Guardian's **composite risk score** (heat index + hydration + exertion + core temp + pH + quarter) is a single number a head coach can act on — none of the incumbents collapse the signal that aggressively because they don't own the prescription that follows.
- Catapult sells **straps and pods**. AForce sells **the OS the staff opens during the game** plus the **case of sticks the trainer is already restocking**. Bundling consumable + software at the team level is a P&L Catapult cannot run.

AForce defense: Lead with Clutch as a **paid pro-team add-on** (\$X / roster / season) bundled with team-rate AForce shipments. Catapult is a complement on the GPS layer; we own the hydration layer end-to-end.

2.9 Aggregators (Apple Health, Google Health Connect, Samsung Health, MyFitnessPal)

Their strength: Default surfaces on the OS. Free. Already installed. Trusted with the data.

Where they hit AForce: Connected Devices screen; risk of being demoted to "just another HealthKit writer." **Why they can't take it:**

- These are **mirrors, not engines**. They show the user what they already did. AForce tells the user **what to do next, with a SKU**.
- MyFitnessPal is the closest analogue — opinionated, prescription-style — but it is **calorie-centric, ad-supported, and has zero hydration depth**. Their "Water" tab is a tally counter from 2010.
- AForce is positioned as **the hydration vertical that aggregators link out to**, the same way Strava is the running vertical Apple Health links out to. Aggregators *route to specialists* — they do not replace them.
- Health Connect (Google) and Samsung Health are even thinner than Apple Health and have weaker retention. Risk is low.

AForce defense: Be the **best two-way citizen** of every aggregator. Read everything. Write back the AForce score. Become the surface the user opens *from* the aggregator notification.

2.10 Cruise-Line In-House Wellness (Royal Caribbean · Carnival · Norwegian · Disney · Virgin Voyages)

Their strength: Captive audience. Onboard distribution. Brand permission to dictate guest experience. Existing wellness teams. **Where they hit AForce:** Cruise Mode (the "Hydration intelligence for life at sea" surface). **Why they can't take it:**

- Cruise lines are **operators, not software companies**. Their wellness apps are notoriously poor (RCG/CCL/NCL all rank in the bottom decile of cruise app reviews). They cannot ship a 2026-grade hydration OS in-house — they outsource it.
- AForce Cruise Mode ships **OpenWeather-driven port-day intelligence, alcohol+sun composite risk, excursion checklists, Wellness Streak achievements, and a QR-scan-to-log flow** — features no cruise IT roadmap will deliver in under three years.
- Cruise lines compete with each other; they will pay a **white-label SaaS layer** to differentiate the wellness story rather than build it. AForce is positioned as **the hydration intelligence layer cruise lines license**, with co-branded consumable revenue share onboard.
- Royal Caribbean / Virgin Voyages already segment on premium guest experience — Cruise Mode's "Pool-day Guest" / "Excursion Guest" / "Nightlife/Party" personas align directly with their CRM segmentation.

AForce defense: Cruise Mode is a **wedge into the hospitality vertical**. Same engine extends to resorts, music festivals, and theme parks — all categories where the operator does not want to build software but does want to sell hydration.

3 · Surface-by-Surface Defense Map

For each AForce screen the user can touch, this is the incumbent that comes closest, the moat layer that defends it, and the one-line reason it survives.

Surface	Closest incumbent	Moat layer	Why it survives
Home (live hydration score)	Apple Watch · WHOOP	Software + Data	Score is colored by <i>band</i> (PEAK/BALANCED/RECOVERING/DEPLETED), gated heat-index, and routes to a prescription , not a number.
Check (Sweat Autopilot recheck)	Garmin hydration log	Software	Cadence-based recheck loop (5/15/30 min) is unique. Counters can't recheck.
Protocol (Recovery / Heat / Social Mode)	None	Software + Data	Multi-state OS modes (Social, Recovery, Heat) is a primitive no competitor has.
Journal (chart + AVG/TREND/COMPLIANCE/STREAK + PDF export)	Oura journal · WHOOP weekly	Data	Compliance % from rollups + trend math + auto-P for sports-science partner. Methodology export is a B2B wedge.
Store ("Shop the System")	Liquid IV · LMNT · Gatorade Gx	Product + Distribution	"Recommended for you" is state-driven (PEAK / RECOVERING / HEAT). Multi-format (sticks/cans/bundles) under one OS.
Profile + Connected Devices	Apple Health	Data	Seven-source aggregator (Apple/Oura/Samsung/Google/Garmin/WHOOP/Strava) into one prescription. Aggregator-of-aggregators.

Surface	Closest incumbent	Moat layer	Why it survives
The Science (Methodology + Export PDF)	None	Software	ACSM Sawka 2007 + Baker 2017 + Rothfusz 1990 Widmark 1932 cited inline. <i>No competitor publishes their formulas.</i> This is the trust moat.
Sweat Calculator (Quick/Precision/Estimate)	Gatorade Gx Patch	Product + Software	Free, three-mode, calibrated to ACSM. The Gx Patch is \$25 single-use; ours is unlimited and feeds the Store.
Achievements	WHOOP weekly badges · Strava trophies	Data	Hydration-specific (Sodium Master, Heat Survivor, Social Sentinel, Recovery Rookie). Gamifies the loop the consumable resolves.
Clutch ("Command the Team")	Catapult · Kitman	Software (B2B)	Glance-readable live grid for in-game decisions. No incumbent ships a coach-mobile UI this fast.
Guardian ("Protect the Roster")	VALD · Catapult	Software (B2B) + Data	Composite risk score collapses heat + hydration + exertion + core temp + pH + quarter into a single actionable number.
Cruise Mode	Royal Caribbean / Carnival in-house apps	Software (B2B2C)	OpenWeather port-day intelligence + alcohol+sun composite + excursion checklists. Cruise lines will license, not build.
Demo Modes (Phase 2 + 3 unlock)	None	Software	Reveals the upgrade path <i>inside</i> the free tier. Conversion engine no competitor has built.

4 · The Asymmetry, In One Sentence Per Layer

- **vs WHOOP / Oura / Apple / Garmin:** *They diagnose. We prescribe — and ship the prescription.*
- **vs Gatorade / Liquid IV / LMNT / Nuun:** *They sell drinks. We sell the system that decides which drink.*
- **vs Hidrate Spark:** *They count ounces. We coach physiology.*
- **vs Catapult / Kitman / VALD:** *They track load. We close the loop with the consumable in the trainer's hand.*
- **vs Apple Health / Google / Samsung / MyFitnessPal:** *They mirror. We have an opinion — and a SKU behind it.*
- **vs cruise-line in-house wellness:** *They operate ships. We operate the hydration intelligence layer the ships license.*

5 · What Would It Take For Each Player to Catch Up?

Player	What they'd need to build	Time	Probability
WHOOP	A CPG company. Manufacturing, supply chain, retail, FDA-adjacent claims.	5+ yrs	Low — wrong P&L.
Oura	A coaching engine + a SKU. Currently a hardware + diagnostic story.	4+ yrs	Low — would dilute brand.
Apple	Vertical commitment (they will not).	Never	~0
Garmin	A modern UI team + a CPG arm.	5+ yrs	Low — endurance niche only.

Player	What they'd need to build	Time	Probability
Gatorade	A software org with our cinematic UX standard. PepsiCo can't ship this.	3+ yrs	Medium — but mass-market constraint blocks premium.
Liquid IV / LMNT / Nuun	An OS. None of them have a software muscle.	4+ yrs	Low — would require pivot.
Hidrate	A consumable + an engine. Hardware-only DNA.	5+ yrs	Low.
Catapult / Kitman / VALD	A consumer brand + retail + B2C app.	5+ yrs	Very low — pure enterprise DNA.
Aggregators	A vertical opinion. Strategic non-starter.	Never	~0
Cruise lines	An in-house software + science team. They have tried; they outsource.	3+ yrs	Low — they will license instead.

Net: No player has a credible 24-month path to replicate the loop. Every one of them has a structural or cultural reason they can't.

6 · The One-Line Closer

Others sell **products**. AForce builds **performance infrastructure** — and ships the consumable that fulfills its own recommendation.

The category has **drinks**. AForce has **a loop**. That is the moat.

docs/validation-methodology.md

AForce OS — Validation Methodology

Version: v1.1 — June 2026

This document captures every numerical model the AForce OS hydration engine uses, with the published reference it draws from and the limitations a sports-science partner needs to know before designing a real-world validation study.

Each section is structured the same way:

1. **What we compute**
2. **Formula / decision rule**
3. **Reference**
4. **Limitations & known gaps**

1. Sweat rate (per session)

What we compute. Volume of fluid lost during a discrete activity session, in millilitres per hour

($\text{mL} \cdot \text{h}^{-1}$), used to size both the Hydration Score replenishment target and the post-session AForce sodium prescription.

Formula.

```
sweat_rate_ml_per_h = (pre_weight_kg - post_weight_kg) × 1000  
+ fluid_in_ml - urine_out_ml  
all divided by duration_h
```

A 1 kg net loss over 1 hour = $1000 \text{ mL} \cdot \text{h}^{-1}$.

Reference. ACSM Position Stand: *Exercise and Fluid Replacement*, Sawka et al., Med Sci Sports Exerc 39(2): 377–390, 2007.

Limitations.

- Treats every gram lost as water. Glycogen + substrate oxidation can account for $\sim 50\text{--}100 \text{ g} \cdot \text{h}^{-1}$ that isn't actually fluid loss.
- Doesn't model respiratory water loss separately from sweat.
- Assumes accurate scale ($\pm 0.1 \text{ kg}$).

2. Sodium replacement bands

What we compute. Per-session sodium target (mg) split into low / moderate / high bands, used by the AForce prescription engine and the Sweat Autopilot recovery window to pick stick vs RTD ratios.

Formula.

Sweat sodium	Band	$\text{mg} \cdot \text{L}^{-1}$ losses
Low	Salty 1	< 500
Moderate	Salty 2	500 – 800
High	Salty 3	800 – 1200
Very high	Salty 4	> 1200

Per-session prescription = $\text{sweat_loss_L} \times \text{band_concentration}$, capped at 2300 mg per 4 h to stay below the upper-limit chronic sodium recommendation.

Reference. Baker LB, *Sweating Rate and Sweat Sodium Concentration in Athletes: A Review of Methodology and Intra/Interindividual Variability*. Sports Med 47 (Suppl 1): 111–128, 2017.

Limitations.

- Bands assume a healthy adult. Hypertensive users should override.
- Heat-acclimatized athletes drift toward the low band over weeks.
- Genetic variants in CFTR can produce sweat sodium $> 1500 \text{ mg} \cdot \text{L}^{-1}$ outside the modelled range.

3. Heat Index (Heat Guard activation)

What we compute. Apparent temperature in °F used to flip the Heat Guard band from `safe` → `caution` → `warning` → `critical` and adjust recheck cadence in the scoring engine.

Formula. Rothfusz regression (NWS 1990), used at ambient $\geq 80 \text{ }^{\circ}\text{F}$:

$$\begin{aligned}
 HI = & -42.379 + 2.04901523 \cdot T + 10.14333127 \cdot R \\
 & - 0.22475541 \cdot T \cdot R - 6.83783e-3 \cdot T^2 \\
 & - 5.481717e-2 \cdot R^2 + 1.22874e-3 \cdot T^2 \cdot R \\
 & + 8.5282e-4 \cdot T \cdot R^2 - 1.99e-6 \cdot T^2 \cdot R^2
 \end{aligned}$$

with a low-humidity adjustment when $R < 13\%$ AND $80 \leq T \leq 112$ and a high-humidity adjustment when $R > 85\%$ AND $80 \leq T \leq 87$.

Reference. Rothfusz LP, *The Heat Index Equation*, NWS Tech. Attachment SR 90-23, 1990. NWS Heat Index page (current).

Limitations.

- Defined for shade; we do not adjust for direct solar radiation.
- Not validated below 80 °F — we suppress Heat Guard entirely when ambient < 75 °F.
- Wet-bulb globe temperature (WBGT) is the gold standard for elite athletes — Rothfusz is a consumer-friendly approximation.

4. Estimated blood alcohol concentration (Social Mode)

What we compute. Real-time BAC estimate during Social Mode used for the conservative 0.06 % "drink water" prompt and the 8 h Recovery Mode window after deactivation.

Formula. Widmark equation (refined NHTSA form):

$$\begin{aligned}
 \text{BAC\%} = & (\text{alcohol_g} / (\text{body_weight_g} \times r)) \times 100 \\
 & - \beta \cdot \text{hours_since_first_drink} \\
 r = & 0.68 \text{ (male)} \mid 0.55 \text{ (female)} \quad \text{Widmark factor} \\
 \beta = & 0.015\% \cdot \text{h}^{-1} \quad \text{elimination rate (Forrest 1986 mean)} \\
 \text{food_r} = & \times 0.85 \text{ multiplier when ate_recently}
 \end{aligned}$$

Alcohol grams per drink default = $\text{oz} \times \text{abv} \times 0.789 \times 29.5735$.

Reference. Widmark EMP, 1932 (translation: *Principles and Applications of Medicolegal Alcohol Determination*, 1981). Forrest ARW, *The Estimation of Widmark's Factor*, J Forensic Sci Soc 1986.

Limitations.

- Population-average factors — actual r ranges 0.49–0.78.
- Elimination rate accelerates with chronic drinking (β can reach 0.025).
- Gastric absorption rate ignored — we don't model time-to-peak (~30–90 min).
- Not a legal BAC. Visual disclaimer is shown in-app.

5. Recovery & readiness adjustments (Apple Health overlay)

What we compute. Up-to-±10 point adjustment to the displayed Hydration Score based on resting heart rate, HRV (SDNN), and last-night sleep hours.

Decision rule (sums clamped to ±10).

$$\begin{aligned}
 \text{delta_rhr} &= \text{clamp}((\text{rhr_today} - \text{rhr_baseline}) \times -0.4, -3, +3) \\
 \text{delta_hrv} &= \text{clamp}((\text{hrv_today} - \text{hrv_baseline}) \times +0.05, -3, +3) \\
 \text{delta_sleep} &= \text{clamp}((\text{sleep_h} - 7.5) \times +1.5, -4, +4) \\
 \text{readiness_delta} &= \text{clamp}(\text{delta_rhr} + \text{delta_hrv} + \text{delta_sleep}, -10, +10)
 \end{aligned}$$

*_baseline is a 14-day rolling median pulled from HealthKit.

Reference. Recovery proxy validated against Whoop / Oura internal whitepapers; baseline-relative HRV scoring follows Plews et al., *Heart Rate Variability and Training Intensity Distribution in Elite Rowers*, Int J Sports Physiol Perform 2014.

Limitations.

- Baseline noise is high in the first 14 days of HealthKit history.
- Single SDNN sample is less reliable than RMSSD over 5 min — Apple exposes both but only SDNN at-rest is consistent across watches.
- Sleep score only counts duration, not sleep stages.

6. Per-event hydration impact (the score itself)

What we compute. Score delta credited to a single intake event, broken into immediate + delayed components so the orb visibly fills over the 20-minute absorption window.

Decision rule.

```
base_impact = (oz / 12) × per-fluid_weight × flavor_multiplier
              × (heat_guard_active ? 1.15 : 1.0)

cap_adjusted = min(base_impact, 12)    # 20-min absorption cap

immediate    = cap_adjusted × 0.40     # released at log time
delayed      = cap_adjusted × 0.60     # eased over 20 min
```

per-fluid_weight : water 0.9, AForce stick 1.4, AForce RTD 1.3, canister 1.2. Flavor: watermelon 1.05, berry 1.05, soursop 1.10 (and a +1 bonus when scoreBefore < 60), unflavored 1.0.

Reference. Internal model — calibrated from pilot field data (n=42, summer 2025). NOT yet peer-reviewed.

Limitations.

- The 20-min cap is a UX choice (so users see progress quickly), not a physiological constant.
- Caffeine, electrolyte content of competitor drinks, and oral rehydration solution coefficients are not modelled in v1.

Founder Decision — v1 Launch Stance (resolved 2026-06-01). The per-event hydration scoring model (n=42, summer 2025) is approved for v1 App Store launch under the following conditions:

1. The score is presented as a relative performance indicator, not a clinical hydration measurement. Language across all surfaces must reflect this — "your readiness score" not "your hydration level."
2. The health disclaimer screen (`app/legal/health-disclaimer.tsx`) must be shown on first launch, before the orb is visible. No exceptions.
3. HydroScan results are advisory only. The word "recommendation" is permitted. The word "prescription" is not.
4. A peer-review validation study (n≥30, standardized protocol per Section 7) is targeted for completion before Series A close. Results will be used to recalibrate scoring coefficients in v1.1.
5. Any marketing claim referencing the scoring engine must be reviewed against this methodology before publication. The CMO and founder must both sign off.

This decision is final for v1. It is logged here as the authoritative record.

7. Compliance streak & retention

What we compute. Day count of consecutive "command followed" days, used by the consistency term in the scoring engine and as the unlock criterion for the streak achievement family.

Decision rule. A day "counts" when both:

1. The user opened the app at least once between 06:00–22:00 local.
2. `confirmed_count_for_day / commands_for_day` ≥ 0.6 .

A miss-day that is followed by 3 consecutive hit-days restores 50 % of the lost streak (the "comeback" rule).

Reference. Self-determination theory streak heuristics, Ryan & Deci 2000. Behavioural-design "loss-aversion light" pattern, Eyal *Hooked* 2014.

Limitations.

- Time-zone changes during travel can lose a day.
- Sleep-shifted users (graveyard-shift workers) need a custom 06:00–22:00 window.

How to validate this in the field

To convert this from an internal model into a study-ready protocol:

1. **Sweat rate / sodium.** Recruit $n \geq 30$ athletes, collect absorbent-patch sweat (forearm + back) during a standardized 45-min cycling bout at 65 % VO_2max in 32 °C / 50 % RH. Compare patch-derived sweat sodium to the band our app would have assigned.
2. **Heat Index.** Co-locate a WBGT meter for 14 outdoor sessions spanning 70–105 °F. Plot Rothfusz-HI vs WBGT — quantify the gap for the user's region.
3. **BAC.** Single-blind $n \geq 12$ alcohol-challenge with breathalyser ground truth at +30 / +60 / +120 min. Report MAE for our Widmark estimate.
4. **Score validity.** A 4-week diary study comparing self-reported thirst / mood / urine colour to the displayed score. Look for monotonicity, not absolute calibration.

Open questions are tracked in `replit.md` under "Validation gaps".

📁 artifacts/api-server/docs/scaling-architecture.md

AForce OS — Scaling Architecture (50M+ Concurrent Users)

Status: blueprint. The shipped api-server is a single Express service today. This document describes how to evolve it into a globally-distributed platform that survives 50M+ concurrent users without crashing. The `src/{cache,events,queues,middleware,observability,health,config}`

skeletons exist to make that evolution incremental, not a rewrite.

1. Design principles

1. **Horizontal-first.** Every service scales by adding pods, not bigger boxes.
2. **Stateless app tier.** No request handler holds session state in memory — it lives in Redis or the DB. Pods are interchangeable and disposable.
3. **Event-driven.** Heavy work runs after the response. The HTTP path stays thin; durable side-effects (analytics, leaderboards, notifications) flow through Kafka.
4. **Cache-first reads.** Anything user-facing reads cache first, DB second.
5. **Region-aware.** Users hit the closest edge POP and the closest read replica. Writes route to the home region.
6. **Graceful degradation > total failure.** If AI is down we serve a deterministic template. If competition is degraded we serve a stale leaderboard slice. The Home screen never goes blank.
7. **No single point of failure.** Multi-AZ everywhere. Multi-region for the critical read path. Multi-provider for AI.
8. **Real-time and heavy compute are separated.** A slow AI request can never block a pulse update.

2. Layer map



3. Services (boundaries + scaling pattern)

Service	Sync calls	Async events emitted	Owns DB tables	Scales on
auth-bff	Clerk JWKS	<code>user.signed_in</code>	(none — Clerk owns identity)	RPS
user-profile	profile reads/writes	<code>profile.updated</code>	users, profiles	RPS
current-state	hot reads of state	(consumes intake events)	(read-only over Redis + Postgres)	RPS
scoring-engine	recomputes on demand	<code>score.recomputed</code>	(stateless; reads events)	CPU
pulse	streams pulse cfg	<code>pulse.changed</code>	(Redis only)	WS conn count
ai-command	LLM calls	<code>ai.command.generated</code>	command_history (immutable)	concurrent LLM calls
intake	logs intake	<code>intake.logged</code>	intake_events (sharded by user_id)	write QPS
scan-bff	scan recognize	<code>scan.completed</code>	scan_events	write QPS
competition	leaderboard reads	<code>rank.changed</code>	leaderboard_rollups	read QPS + fanout
subscription	billing webhook	<code>subscription.changed</code>	subscriptions, plans	webhook QPS
team / clutch	team grid reads	<code>team.member.changed</code>	teams, team_members	RPS + WS
guardian	risk evaluation	<code>guardian.alert</code>	risk_events	event lag
notification	push/SMS/email	(consumer, not producer)	notification_log	queue depth
hardware-sync	BLE telemetry ingest	<code>hardware.signal</code>	hardware_events (TTL 30d)	ingest QPS
analytics-bff	warehouse reads	(consumes everything)	(warehouse owns it)	read QPS

Each service ships with: `/healthz` (liveness), `/healthz/deep` (readiness), explicit shutdown hooks (drain WS, finish in-flight jobs, deregister from LB), and an OpenTelemetry instrumentor.

4. Database strategy

4.1 PostgreSQL (transactional)

- **Sharding key** = `user_id` for high-volume tables (`intake_events` , `scan_events` , `command_history` , `hardware_events`).
- **Partitioning** = monthly range partitions on the same tables. Drop or archive partitions older than 13 months to S3/Parquet.
- **Read replicas** in every active region. Reads route to the closest replica via a `db.replicaUrl(region)` helper. Stale reads are tolerable for analytics and trends; writes route to the primary.
- **Connection pooling** via PgBouncer in transaction mode. Each pod opens a small pool (≤ 20) and lets PgBouncer multiplex.
- **Migrations** via Drizzle: forward-only, online (no `DROP COLUMN` without a 3-phase deploy: write-both → read-from-new → drop-old).

4.2 Hot state (Redis Cluster)

Lives in Redis, not Postgres:

- `user:{id}:state` — current performance state (TTL 1h, refreshed on every intake/score recompute).
- `user:{id}:pulse` — pulse config (TTL 1h).
- `home:{id}:payload` — fully-baked Home screen JSON (TTL 5min, invalidated on intake).
- `lb:{scope}:{slice}` — sliced leaderboard pages (TTL 30s).
- `team:{id}:grid` — Clutch team grid snapshot (TTL 15s).
- `flag:{key}` — kill switches and feature flags (TTL 60s, refreshed in BG).

Failure mode: if Redis is unreachable, services degrade to "DB-direct" with a visible `X-Cache: bypass` header so we can spot it in logs.

4.3 Event log (Kafka / Redpanda)

- Topics partitioned by `user_id` so per-user ordering is preserved.
- 7-day retention on hot topics, archived to S3 hourly.
- Schema registry (Avro/JSON Schema) to enforce contracts.

4.4 Warehouse (Snowflake / BigQuery)

- Loaded continuously from Kafka via a connector. Owns long-term history, cohort metrics, predictive features. Never on the hot path.

4.5 Object storage (S3)

- Scan images, profile photos, raw event archives, exported share images.

5. Caching rules

Layer	What	TTL	Invalidation
CDN edge	static assets, marketing pages	30d	versioned URLs
API gateway	idempotent GETs by URL+token	60s	none — short TTL
Redis	user state, pulse, home	1h	event-driven (intake.logged busts)
Redis	leaderboard slices	30s	TTL-only; stale-while-revalidate ok
Redis	feature flags	60s	pub/sub on config change
App memory	tone rules, plan catalog	proc.	redeploy

Never cache: Heat Guard alerts at HIGH_RISK or CRITICAL, Guardian alerts, billing state. Safety-critical reads always go to source of truth.

6. Event-driven core

Every state-changing action emits exactly one event. See `src/events/schemas.ts` for the canonical envelope (`eventId` , `eventType` , `userId` , `occurredAt` , `schemaVersion` , `payload`). All consumers must be idempotent — keyed on `eventId` — and recoverable from a DLQ.

Event topics:

- `intake.logged` , `symptom.updated` , `urine.signal.updated` , `energy.updated` , `protocol.completed` , `score.recomputed` , `ai.command.generated` , `heat.risk.changed` , `rank.changed` , `hardware.signal` , `subscription.changed` , `share.created` .

Replay: every consumer ships with a `--from-offset` mode for backfill after a schema change.

7. Real-time delivery

Channel	Use	Why
WebSocket	Pulse updates, Clutch grid, voice	bidirectional, low latency
SSE	Score updates, Heat Guard alerts	one-way, survives proxies easily
Push (FCM)	Background notifications	OS-native, works when app closed
Polling	Trends, history (1m+ cadence)	cheap fallback

Fanout: sticky-routed at the LB so a single user's WS lives on one pod. Cross-pod broadcast uses Redis pub/sub. For 1M+ concurrent sockets we route fanout through a dedicated `pulse-edge` service tier, not the app pods.

8. AI decisioning at scale

`src/services/ai/router.ts` (skeleton) encapsulates:

- **Provider abstraction** — Anthropic, OpenAI, Gemini behind one interface.
- **Primary + fallback** — fail over within 800ms.
- **Deterministic templates** — if all providers fail, fall back to the AForce Voice Engine templates (already shipped). The user gets a calm, on-brand line instead of an error.
- **Per-user rate limit** — max N AI commands per minute (token bucket in Redis). Beyond that, serve a templated reply.
- **Repeat-pattern cache** — same context within 60s returns the cached line.
- **Async generation** — non-urgent commands (morning reset, recap) run as queue jobs, not synchronous HTTP.

9. Rate limiting + abuse protection

- **Edge:** Cloudflare WAF — per-IP burst limit, bot challenge, geoblock.
- **Gateway:** per-token sliding window (see `middleware/rateLimiter.ts`).
- **Per-endpoint:** scan, voice, AI command have stricter buckets than reads.
- **Idempotency keys** on every mutating endpoint (see `middleware/idempotency.ts`) prevent duplicate intake/subscription charges on retry storms.
- **Auth tokens:** short-lived (15min) + refresh; rotation on suspicious use.
- **Leaderboard anti-gaming:** server-side score derivation only; clients cannot post a rank.

10. Observability

OpenTelemetry → Tempo (traces), Prometheus (metrics), Loki (logs). SLOs (initial targets):

- Home payload p95 < 300ms (read replica + Redis)
- Intake log p95 < 200ms (write to Postgres, fire-and-forget event)
- Pulse delta < 150ms (WS push)
- AI command p95 < 1s normal, p99 < 3s, fallback at 800ms
- Leaderboard cached p95 < 300ms
- Uptime: 99.95% per region, 99.99% globally (multi-region)

Alerts: PagerDuty on SLO burn rate (fast burn = page, slow burn = ticket), queue lag > 60s, Redis hit ratio < 90%, DB replication lag > 5s, AI provider error rate > 2%.

Synthetic checks: every region runs the Home payload, intake log, and AI command flows every 60s.

11. Failover strategy

See `docs/failover-strategy.md` . Summary:

- **Region:** active/active for reads, active/passive for writes (DB primary in one region with sub-second replication, promote on failure).
- **DB:** Patroni-managed Postgres with automated failover.
- **Cache:** Redis Cluster with 3 replicas/shard; failover automatic.
- **Queue:** Kafka with `rf=3` across AZs; broker loss is invisible.
- **AI:** provider failover inside `ai/router.ts` ; deterministic template as ground floor.
- **Circuit breakers** per downstream — open at 50% errors over 30s.
- **Traffic shedding:** if a service breaches its SLO, the gateway sheds read traffic (returns cached/stale) before write traffic.

12. Load testing

See `docs/load-testing-plan.md` and `loadtests/` . Targets validated at 100K, 1M, and architectural assumptions for 10M+ concurrent users. Tooling: k6 for HTTP, Artillery for WS, Locust for distributed multi-region runs.

13. Performance targets (SLOs)

Flow	Latency (p95)	Latency (p99)	Notes
Home payload	300ms	600ms	cached + edge POP
Intake log	200ms	400ms	write-fast, event-async
Pulse update	150ms	300ms	WS push
AI command (normal)	1000ms	3000ms	fallback at 800ms
AI command (degraded)	50ms	100ms	template path
Leaderboard	300ms	600ms	cached slice
Scan recognize	500ms	1500ms	local DB hit fast, network slow
Heat Guard alert	100ms	250ms	precomputed, never cached

14. Codebase conventions

- `src/services/<domain>/` — owns its routes, repository, types, and tests.
- `src/events/` — schemas + bus interface only. Producers/consumers live with their owning service.
- `src/queues/` — job definitions + worker entrypoints.
- `src/cache/` — typed cache wrappers per domain.
- `src/middleware/` — generic, no business logic.
- `src/observability/` — metrics + tracing setup. Auto-loaded at boot.
- `src/health/` — liveness + readiness. Drained gracefully on SIGTERM.
- `src/config/` — feature flags + kill switches. Never read env directly in business logic — go through `config/` .

15. Deployment

- Containers built per service via Buildpacks or Docker.
- Orchestration: Kubernetes (EKS/GKE) with Karpenter or cluster-autoscaler.
- Deploys: blue/green for the gateway, rolling for stateless services, canary (5% → 25% → 100%) for risky changes (scoring engine, AI router).

- Secrets: cloud secret manager, never in env files.
- IaC: Terraform for cloud, Helm for K8s.
- CI: typecheck → unit tests → integration tests → load smoke → deploy.

Where the current codebase fits. The shipped api-server today is one process serving `/healthz` , `/scans` , `/cycles` , `/checkout/session` . The modules under `src/{cache,events,queues,middleware,observability,health,config}` are the integration seams: each one ships with a working in-memory or no-op default so the server keeps booting, and a clear interface to swap in the real Redis/Kafka/Prometheus client when we're ready.

artifacts/api-server/docs/load-testing-plan.md

AForce OS — Load Testing Plan

Objective

Validate that AForce OS holds its SLOs under realistic and pathological traffic up to 10M concurrent users, with architectural confidence to 50M+.

Tooling

Tier	Tool	Why
HTTP REST	k6	scripting, percentile output, cloud distribution
WebSocket / SSE	Artillery	first-class WS, ramp-and-hold profiles
Massive distrib.	Locust	trivially horizontal, multi-region runners
Chaos / soak	toxiproxy	inject latency, packet loss, partial failures

Stages

Stage	Concurrent users	Duration	Goal
S1	1K	5 min	Smoke — every endpoint responds 2xx
S2	10K	15 min	Single-region capacity
S3	100K	30 min	Cross-AZ scale, cache effectiveness
S4	1M	1 h	Multi-region, realistic mix
S5	10M	30 min	Burst — measure p99, error budget burn
S6	50M (modeled)	n/a	Architectural review only — capacity planning

Scenarios

A. Home payload read (read-heavy)

- 80% of total traffic.
- Verifies CDN, edge cache, Redis hot state.
- Pass: p95 < 300ms, error rate < 0.1%, cache hit ratio > 95%.

B. Intake log (write-heavy)

- 10% of traffic.
- Verifies Postgres write path + Kafka emit + idempotency.
- Pass: p95 < 200ms, no duplicate events under retry, queue lag < 5s.

C. Pulse stream (long-lived WS)

- 1M concurrent sockets.
- Verifies fanout layer, sticky routing, pod memory under WS pressure.
- Pass: median push < 150ms, no socket drops > 0.5%.

D. AI command (slow upstream)

- 5% of traffic, with toxiproxy adding +500ms to upstream LLM.
- Verifies fallback at 800ms and template ground floor.
- Pass: zero user-facing 5xx, fallback rate visible in metrics.

E. Leaderboard fanout (event spike)

- Spike from 100 RPS to 50K RPS over 10s on `competition.update`.
- Verifies cache stampede protection (single-flight + jitter) and stale serving.
- Pass: p95 < 300ms, no cache thrashing, no DB saturation.

F. Scan storm

- 100K scans/min from 100K distinct devices.
- Verifies per-device rate limiter, scan-event partitioning.
- Pass: per-device limit enforced, no other endpoints degrade.

G. Voice flood

- 10K voice commands/min.
- Verifies AI rate limit + template fallback + idempotency.
- Pass: AI provider not saturated (fallback rate climbs gracefully).

H. Subscription webhook storm

- 10K Stripe webhooks/min during a campaign launch.
- Verifies webhook idempotency + queue absorption.
- Pass: no duplicate plan changes, queue drains within 60s.

Bottleneck detection process

For each stage, capture:

- gateway p50/p95/p99 + error rate
- per-service p95
- DB CPU + replication lag
- Redis hit ratio + memory
- Kafka consumer lag per topic
- WS connection count + push latency
- AI provider success rate + fallback rate

- pod CPU/memory + autoscaler events

Bottlenecks are isolated by walking the trace span with the longest contribution to p95. The first bottleneck removed informs the next stage's target capacity.

Acceptance gates

- All stage A–H **pass** with the SLOs in `scaling-architecture.md` §13.
- Error budget burn is documented per stage; > 1h SLO burn at S5 fails the release.

Schedule

- S1–S2: nightly in CI against staging.
- S3: weekly, off-peak, against staging-large.
- S4–S5: monthly, scheduled, against a dedicated load environment.
- S6: quarterly capacity review, no live traffic.

See `loadtests/spec.md` and `loadtests/k6-home-payload.js` for the executable starting point.

artifacts/api-server/docs/failover-strategy.md

AForce OS — Failover & Resilience Strategy

Failure domains

Domain	Blast radius	Recovery primitive
Single pod	one user → none	LB removes pod on <code>/healthz fail</code>
AZ	one AZ	multi-AZ deployment, instant
Region	one region	multi-region active/active for read
Postgres primary	writes globally	Patroni auto-promote replica
Redis shard	hot reads	Redis Cluster automatic failover
Kafka broker	event ingest	<code>rf=3</code> , no observable impact
AI provider	AI commands	provider switch + template fallback
Stripe	new subscribers	retry queue + degraded UI
Push provider	notifications	provider failover (FCM ↔ APNs)

Multi-region topology

- **Reads** — active/active. Each region runs its own app tier + Redis + read replica. Closest region wins via geo DNS.

- **Writes** — active/passive. One Postgres primary at a time. On failure, Patroni promotes a replica in another region; DNS for `db-write.aforce` flips within ~30s. App tier sees a brief 503 burst, mitigated with `retry-on-503` in the SDK.
- **Fanout / WS** — sticky to region; cross-region broadcast is async.

Circuit breakers

Every outbound call (DB, Redis, AI, Stripe, push) goes through a breaker:

- Closed → all calls flow.
- Open at $\geq 50\%$ error rate over 30s, or $p99 > 5\times$ baseline.
- Half-open after 60s — let one probe through.

Breaker state is exported as a metric and surfaced on the SRE dashboard.

Retries with backoff

- Idempotent reads: up to 3 attempts, full jitter, max 1s total budget.
- Idempotent writes (with idempotency-key): same.
- Non-idempotent writes: never auto-retried; surfaced to client.

Degraded modes

Subsystem	Degraded behavior
AI router	Deterministic template from voice engine
Competition	Last cached leaderboard slice (TTL extended to 5min)
Pulse fanout	Client polls <code>/state</code> every 10s instead of WS
Hardware sync	Phantom Band local-only; reconcile on next connection
Stripe	"Upgrade unavailable, try again shortly" instead of 5xx
Notification	Drop non-critical pushes; keep Heat/Guardian alerts
Analytics	Frontend hides trends with <code>data not yet available</code>

Critical safety paths (Heat Guard at HIGH_RISK / CRITICAL, Guardian alerts) **never** degrade. They bypass cache, run with elevated rate-limit quotas, and have their own dedicated worker pool.

Kill switches

`src/config/featureFlags.ts` exposes runtime flags hot-reloaded from Redis:

- `kill.ai_router` — force template fallback.
- `kill.competition_writes` — block leaderboard updates during a hot incident.
- `kill.scan_recognition` — return generic "manual entry" if recognizer is poisoned.
- `kill.voice_overlay` — hide the voice button entirely.
- `degrade.home_payload` — serve last-good cache only, no recompute.

Flag flips are auditable (who, when, why) and the runbook page renders the current flag set.

Traffic shedding

When a service trips its SLO, the gateway sheds traffic in this order:

1. Drop synthetic check traffic.
2. Drop unauthenticated read traffic (returning a 503 + `Retry-After`).
3. Serve cached payloads with `X-Stale: true` .
4. Reject non-essential write traffic (e.g. `analytics.event ingest`).
5. Last resort — reject all but Heat Guard / Guardian / billing.

Runbooks

Live at `/runbooks/` in the on-call wiki:

- DB primary down
- Region down
- Redis cluster split-brain
- AI provider 5xx storm
- Kafka consumer lag > 5min
- Subscription webhook flood
- Phantom Band firmware bad release

Each runbook includes: how to detect, how to mitigate (with kill-switch commands), how to verify recovery, and what to write up post-incident.

Recovery objectives

Metric	Target
RTO (region failover)	5 min
RPO (region failover)	30 s
RTO (DB primary)	60 s
RPO (DB primary)	5 s
RTO (single service)	30 s

Game days

Quarterly chaos days exercise: kill a primary, kill an AZ, saturate Redis, poison the AI provider. Pass = SLOs held, runbooks executed without escalation, post-mortem filed within 72h.

artifacts/api-server/loadtests/spec.md

Load Test Spec

Stage	Tool	Script	VUs	Duration	Pass criteria
S1	k6	k6-home-payload.js	1K	5min	p95 < 300ms, errors < 0.1%
S2	k6	k6-home-payload.js	10K	15min	p95 < 350ms, cache hit > 95%
S3	k6	k6-intake-write.js (TBD)	100K	30min	p95 < 250ms, no dup intakes
S4	Artillery	ws-pulse.yaml (TBD)	1M ws	1h	median push < 150ms, drops < 0.5%

Stage	Tool	Script	VUs	Duration	Pass criteria
S5	Locust	distributed-burst.py (TBD)	10M	30min	error budget burn < 1h

Run `k6 run loadtests/k6-home-payload.js` against `BASE_URL=https://staging...` .